

White
Paper

ENTERPRISE  NTEGRATION, INC. / IDS SCHEER 

ARIS for DoDAF

ARIS for DoDAF

ARIS for DoDAF

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Executive Summary

The DoD Architecture Framework, Version 1.0, defines a common approach for Department of Defense (DoD) architecture description development, presentation, and integration. The Framework is intended to ensure that architecture descriptions can be compared and related across organizational boundaries, including Joint and multinational boundaries.

The Framework defines three related views of architecture: operational (OV), systems (SV), and technical standards (TV). Each view is composed of sets of architecture information that are depicted via graphic, tabular, or textual products. The All-DoD Core Architecture Data Model (CADM) defines the data structure and relationship for architecture information.

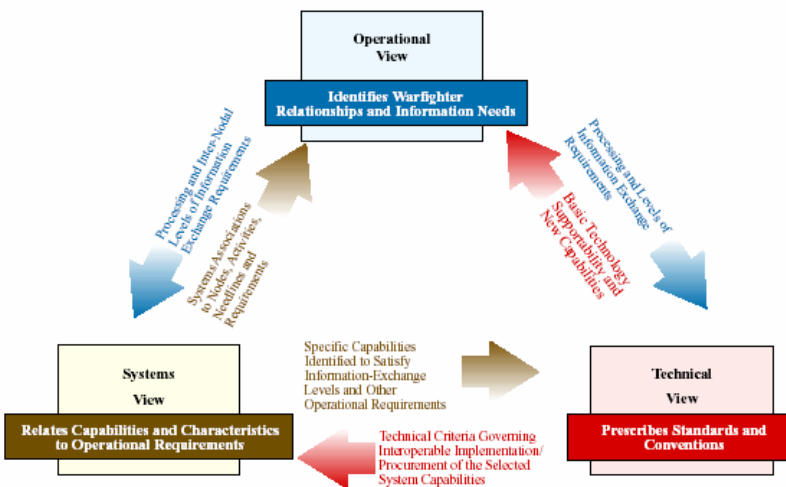


Table: The Three DoDAF Architecture Views

In June 1996 the C4ISR Architecture Framework, Version 1.0, was developed as a product of the Integrated Architecture Panel (IAP), which was chartered by the Assistant Secretary of Defense for Command, Control, Communications, and Intelligence (ASD[C3I]). In October of that year, the C4ISR Architecture Working Group (AWG) was established by the Principal Deputy Assistant Secretary of Defense (PDASD) for C3I and the Joint Staff/J6. The AWG refined the framework and published the C4ISR Architecture Framework, Version 2.0 in December 1997.

The usefulness of the C4ISR Architecture Framework, combined with both Federal and DoD policy encouraging the use of enterprise architectures, led the DoD to revise the document, resulting in the DoD Architecture Framework, Version 1.0, which was approved in 2003. The framework is the work of representatives of the Joint Staff, the Military Services, and various OSD components.

The DoDAF defines “what” should be included in an architecture description. As it is designed to be a flexible architectural framework, the DoDAF does not define “how” the architecture should be described and also does not define or limit the supporting tools and notations used to build a DoDAF compliant architecture.

Our main objective in providing DoDAF tool support was to offer a solution that allows a user to have all DoDAF products in one database and all products and their data elements fully integrated. Only by having all products integrated will you be able to see the impact of the system architecture on the operation architecture and vice versa.

In this paper, we will discuss architectures and frameworks in general terms, then describe the ARIS Framework and DoDAF in more detail. The heart of the document is the mapping between the frameworks that shows how ARIS can support DoDAF users. Finally, we will briefly discuss a high-level process for architecture construction.

Architectures and Frameworks

The Business Architecture Concept

In order to describe an enterprise, it is useful to first start with an appropriate **business framework**. The business framework provides a concept to describe business objects and their relationships. It is essential for the success of any business initiative to select a framework that contains the appropriate business objects and offers sufficient methods to describe their characteristics and relationships. The DoDAF is an example of a framework, as is the ARIS framework. As long as the frameworks contain the same information, they can be synchronized. “ARIS for DoDAF” is such a product. It makes use of the power of the ARIS Framework while capturing the data and relationships necessary to show DoDAF compliance.

An **enterprise business architecture** is a customer-specific instance of a business framework. The enterprise business architecture follows the conceptual rules set forth in the parent framework, but is populated with information specific to the organization that will be using it.

A **business process architecture** is documentation or a depiction of the arrangement of business processes within an organization. This arrangement includes connectivity between processes (process flow) and segmentation according to level of detail. The business process architecture is a subset of the enterprise business architecture, it is just one dimension. A complete enterprise business architecture also contains information about products, systems, organizations, and data.

This complete architecture must be captured in way that makes the relationships within and among the different dimensions easy to understand, and that is where an architecture **tool**¹ becomes useful. The tool must support the selected framework and incorporate a sound **method**². A modeling method is a standardized way of describing reality. It will define the model and object types used by the tool, and as such, defines a **meta architecture**, which underlies the process, system and data architectures. This meta architecture defines the modeling components and how they can be organized to describe the enterprise business architecture. In other words, it is a procedure-independent definition of model types, object types, relationships used, modeling rules and attribute types that ensures a consistent modeling approach.

The “output” of a method is a set of **models**. The models, based on common **model types** – templates providing an abstract system of notations that characterize objects and the relationships between them –

¹ Standard frameworks and methods are usually supported by a variety of tools. Some also have tools specifically created for them. For example, the ARIS Toolset was originally developed to support the ARIS framework.

² Models can be evaluated by using general principles of modeling (See Scheer, Business Process Frameworks, Springer, 1998). The method selected to describe the business architecture should be evaluated using these principles within the context of the specific project goals.

are the “user interface” to architectural information. A good, hierarchical³ set of models allows the people in the organization to see important information about the enterprise in a way that is easy to understand.

The architecture should be completed with a set of **documentation**. The documentation can include a variety of additional information that can be connected with the business models. It should also include procedures that tell the users how to use the enterprise business architecture.

ARIS Architecture and Framework

ARIS Architecture

The ARIS architecture is a Three-tier Architecture of Business Process Excellence. The architecture is built on three layers: the strategy layer, the process specification layer, and the execution layer.

At the strategy level, aspects like the general business process structure and strategy, planned innovation and the underlying applications system architecture are defined (Scheer, Abolhassan, Jost, Kirchmer; 2003). The strategy is passed down to the process specification layer where it is transformed into a business process model. That model is used on the execution layer to implement processes, applications, and workflows. The activities on the execution level are monitored and feedback is provided for the controlling activity on the specification level.

³ Advanced modeling tools allow a user to create a model hierarchy. A model hierarchy is a logical structure that creates relationships between different models that exist at different levels of detail or organization. Decomposing objects of one model into smaller entities, which are components of a new model, creates such a hierarchy. A model hierarchy can be created using one or more strategies; for example, function-oriented decomposition, a decomposition of a system into its component applications, etc. Model hierarchies can contain many models and model types in multiple levels and dimensions.

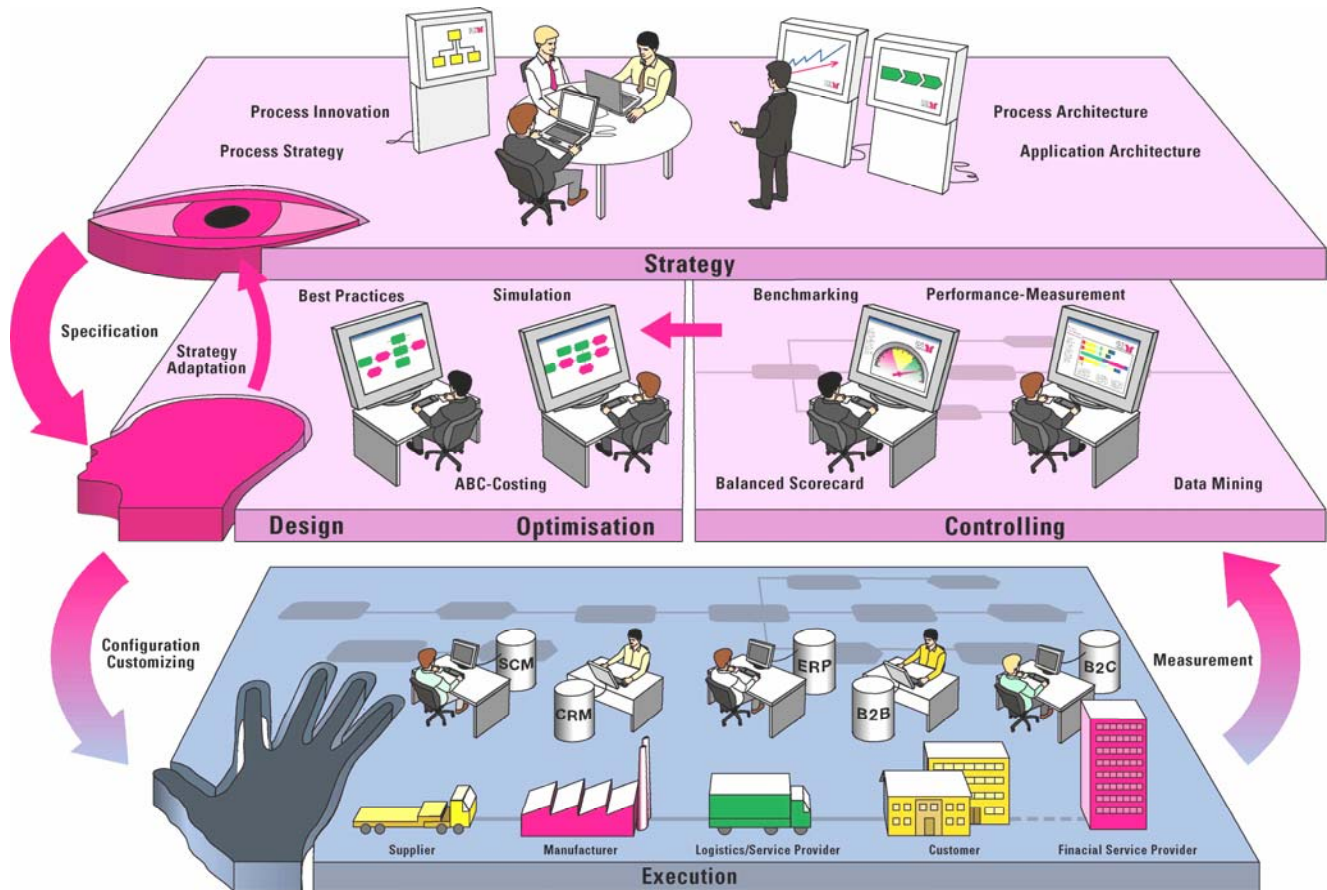


Figure: Three-Tier Architecture of Business Process Excellence

ARIS Framework

To represent processes clearly and systematically, ARIS framework includes four “static” views:

The **product/service view** structures all material and non-material inputs and outputs that are brought into, or performed by, the business process.

Information objects and their attributes, as well as the relationships between information objects are described in the **data view**. Because events represent the status of the process at a certain point in time, they are also assigned to the data view.

The transactions that transform performances and the static relationships between them are described in the **function view**. The terms “function,” “transaction” and “activity” are used here synonymously. Application systems are also included in the function view, because they determine the computer-supported processing rules for activities.

Organizational elements and their relationships form the organizational structure and are described in the **organization view**. In addition to these human resources, operating resources and computer hardware are also assigned to this view.

These views are static views, in that while they may *imply* a process flow, they do not *explicitly show* a process flow⁴. Consequently, the information captured in the static views must be combined into the dynamic process view. The **process view** (or **control view**) displays the static connections between the objects of the data, function, performance and organization views, *and* the dynamic, chronological process flows.

All five views are combined into a construct known as the “ARIS House”.

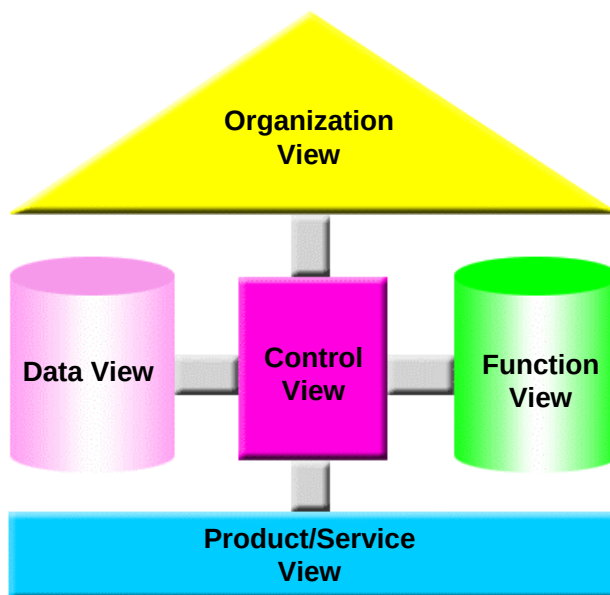


Figure: ARIS House representing the ARIS Framework

Each of the ARIS views is supported by many model types. Due to the large number, we will not explain them all here (the model types most germane to DoDAF will be explained in a later section).⁵ However, an example is the *extended event-driven process chain (eEPC)*. The eEPC is the most often used model type in the process view. Using extended event-driven process chains, you can model the procedural organization of your company, i.e. the procedural sequence of functions within the scope of individual business processes. For this purpose, ARIS provides you with two possible approaches. You can elect a "**lean**"

⁴ This is an important distinction. For example, many organizations attempt to document their business process flows in formats like IDEF0. However, while IDEF0 does show information flowing between activities, it does not explicitly give information on the sequencing of those activities – the sequencing is critical for business process engineering.

⁵ For a complete description of the model types available in ARIS, see *the ARIS Method Manual*.

modeling approach, i.e. a purely **chronological procedural sequence** of process steps (i.e. functions). Or, you can integrate objects of **static** views in this model type (data, function, performance, and organization views) and thereby model the processes **in detail**.

The advantage of dividing the initial problem into individual views is that the complexity of the business processes is reduced. One can look at one view without being overwhelmed with information from the other views. However, the interrelationships created between organizational, data-technical, performance-related, and functional aspects when designing the views are stored and maintained so the components of the different views can be reassembled in the process view when the whole picture is needed.

The next figure is an example of how the ARIS toolset allows you to take data from the static views and make it available in the context of the business process flow. As you can see, each view is supported by a model type and the information from the static models is available in the central eEPC (in the control view box).

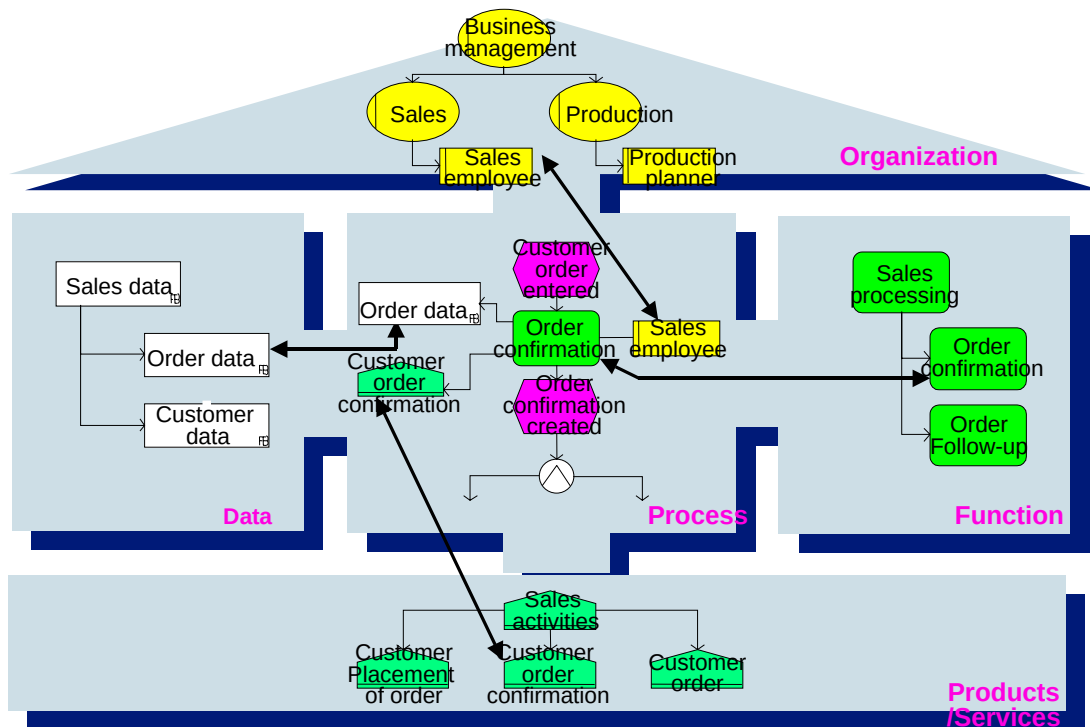


Figure: ARIS House with Model Examples

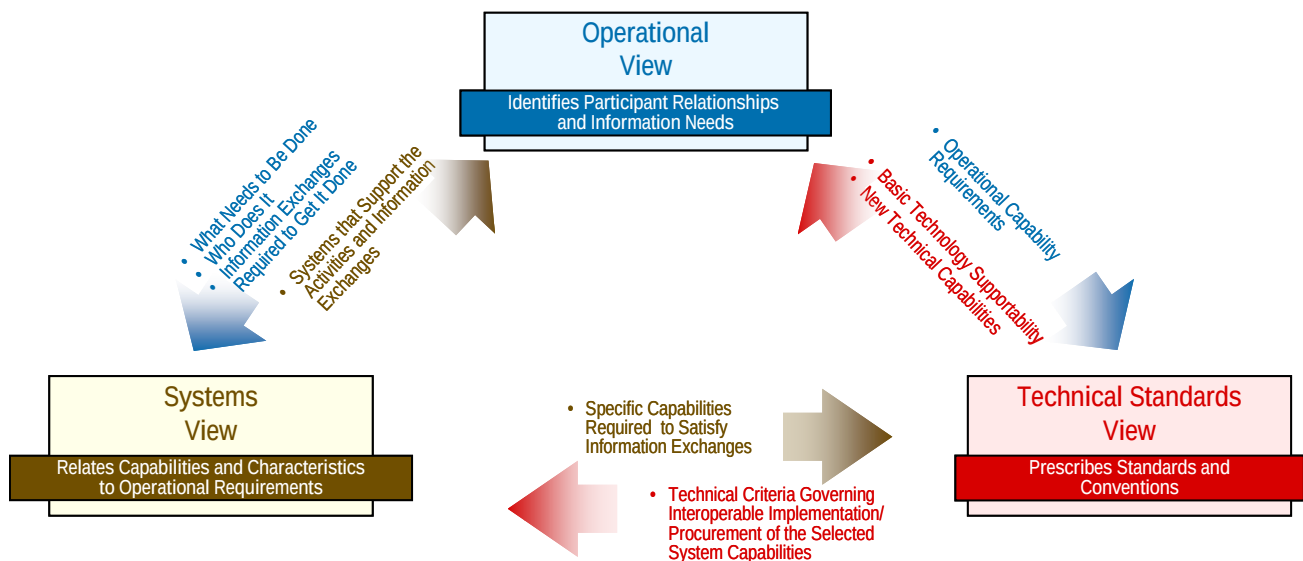
The Department of Defense Architecture Framework (DoDAF)

For the purposes of this white paper, we assume the reader is already somewhat familiar with the DoDAF. However, we will briefly introduce it here in order to set the stage for the description of ARIS for DoDAF.

In June 1996 the C4ISR Architecture Framework, Version 1.0, was developed as a product of the Integrated Architecture Panel (IAP), which was chartered by the Assistant Secretary of Defense for Command, Control, Communications, and Intelligence (ASD[C3I]). In October of that year, the C4ISR Architecture Working Group (AWG) was established by the Principal Deputy Assistant Secretary of Defense (PDASD) for C3I and the Joint Staff/J6. The AWG refined the framework and published the C4ISR Architecture Framework, Version 2.0 in December 1997.

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Like the ARIS Framework, the DoD Architecture Framework, describes architectures through “views”. In DoDAF, there are three views: operational (OV), systems (SV), and technical standards (TV). Each view comprises several architecture products⁶ (akin to the model types in the ARIS Framework) and these products are interrelated both within and across views. Underlying the framework is the All-DoD Core Architecture Data Model (CADM), which is an attempt to define a standard data structure and the relationships between architecture data elements. The views are depicted in the next figure:



Three views of DoDAF (DoDAF, v1.0)

⁶ The DoDAF defines architecture products as: “those graphical, textual, and tabular items that are developed in the course of gathering architecture data, identifying their composition into related architecture components or composites, and modeling the relationships among those composites to describe characteristics pertinent to the architecture’s purpose.” (DoDAF v1.0)

The products in each of the views are listed below. In the next section, we will describe each product in more detail, along with the ARIS model types that can be used to represent each product.

Operational View Products		
View	Name	Description
OV-1	High-Level Operational Concept Graphic	High-level graphical/ textual description of operational concept
OV-2	Operational Node Connectivity Description	Operational nodes, operational activities performed at each node, connectivity and information exchange needlines between nodes
OV-3	Operational Information Exchange Matrix	Information exchanged between nodes and the relevant attributes of that exchange
OV-4	Organizational Relationships Chart	Organizational, role, or other relationships among organizations
OV-5	Operational Activity Model	Operational Activities, relationships among activities, inputs and outputs. Overlays can show cost, performing nodes, or other pertinent information.
OV-6a	Operational Rules Model	One of the three products used to describe operational activity sequence and timing - identifies business rules that constrain operation
OV-6b	Operational State Transition Description	One of three products used to describe operational activity sequence and timing - identifies business process responses to events
OV-6c	Operational Event-Trace Description	One of three products used to describe operational activity sequence and timing - traces actions in a scenario or sequence of events and specifies timing of events
OV-7	Logical Data Model	Documentation of the data requirements and structural business process rules of the Operational View.

Systems View Products		
SV-1	Systems Interface Description	Identification of systems and system components and their interconnections, within and between nodes
SV-2	Systems Communications Description	Systems nodes and their related communications lay-downs
SV-3	Systems-Systems Matrix	Relationships among systems in a given architecture; can be designed to show relationships of interest, e.g., system-type interfaces, planned vs. existing interfaces, etc.
SV-4	Systems Functionality Description	Functions performed by systems and the information flow among system functions
SV-5	Operational Activity to Systems Function Traceability Matrix	Mapping of systems back to operational capabilities or of system functions back to operational activities
SV-6	Systems Data Exchange Matrix	Provides details of systems data being exchanged between systems
SV-7	Systems Performance Parameters Matrix	Performance characteristics of each system(s) hardware and software elements, for the appropriate timeframe(s)
SV-8	Systems Evolution Description	Planned incremental steps toward migrating a suite of systems to a more efficient suite, or toward evolving a current system to a future implementation
SV-9	Systems Technology Forecast	Emerging technologies and software/hardware products that are expected to be available in a given set of timeframes, and that will affect future development of the architecture
SV-10a	Systems Rules Model	One of three products used to describe systems activity sequence and timing— Constraints that are imposed on systems functionality due to some aspect of systems design or implementation

SV-10b	Systems State Transition Description	One of three products used to describe systems activity sequence and timing— Responses of a system to events
SV-10c	Systems Event-Trace Description	One of three products used to describe systems activity sequence and timing -- System-specific refinements of critical sequences of events and the timing of these events
SV-11	Physical Schema	Physical implementation of the information of the Logical Data Model, e.g., message formats, file structures, physical schema

Technical View Products		
TV-1	Technical Standards Profile	Extraction of standards that apply to the given architecture
TV-2	Technical Standards Forecast	Description of emerging standards that are expected to apply to the given architecture, within an appropriate set of timeframes

In addition to the three views that describe an architecture in DoDAF, there are two “all-view” (AV) products that relate to the other views but do not provide a distinct “picture” of a portion of the architecture.

AV-1	Overview and Summary Information	Scope, purpose, intended users, environment depicted, analytical findings
AV-2	Integrated Dictionary	Data repository with definitions of all terms used in all products

ARIS for DoDAF

The DoDAF defines “what” should be included in an architecture description. As it is designed to be a flexible architectural framework, the DoDAF framework does not define “how” the architecture should be described and also does not define or limit the supporting tools and notations to be used to build a DoDAF compliant architecture.

Our main objective in providing DoDAF tool support was to offer a solution that allows a user to have all DoDAF products in one database and all products and their data elements fully integrated. Only by having all products integrated will you be able to see the impact of the system architecture on the operation architecture and vice versa.

This section describes in detail how ARIS enables an integrated DoDAF architecture. It presents each DoDAF product description and shows an ARIS model type that can be used to represent the product. A summary of the mapping can be found in the following table. It is important to note that several of the DoDAF products are text products rather than “models”, per se. As such, they are better developed in a tool like a word processor. The document can then be linked into the ARIS database (through a technical terms model, for example) wherever appropriate to make it available to the user. Other products can be handled as either models or as reports generated from the information underlying the models. Before proceeding with the detailed analysis, keep in mind that the standard ARIS product provides more than 150 model types and notations that can answer many kinds of architectural requirements. For example all UML notations are available in ARIS as well as standard notions like organizational charts, entity relationship diagrams or functions trees. All the information captured in the different views in ARIS is fully integrated in one object-linked repository. What we are presenting is a minimum set, in a sense – the simplest mapping between DoDAF products and ARIS model types. It is quite likely that the user may choose to expand the model selection based on his specific needs.

DoDAF View	ARIS View
AV-1	Technical Terms Model
AV-2	Technical Terms Model
OV-1	Value-Added Chain Model (VACD) Event-Driven Process Chain Model (EPC)
OV-2	Value-Added Chain Model (VACD) Event-Driven Process Chain Model (EPC)

OV-3	IE Data Model (IEDM) Input/Output Diagram (IOD)
OV-4	Organizational Chart Value-Added Chain Model (VACD)
OV-5	Function Tree Value-Added Chain Model (VACD)
OV-6	Event-Driven Process Chain Model (EPC)
OV-7	Entity Relationship Model (ERM)
SV-1	Application System Diagram Access Diagram (physical)
SV-2	Access Diagram (physical) Network Diagram
SV-3	Access Diagram (physical) IE Data Model (IEDM)
SV-4	Function Tree Event-Driven Process Chain Model (EPC)
SV-5	*
SV-6	IE Data Model (IEDM)
SV-7	Key Performance Indicator Allocation Diagram
SV-8	*
SV-9	*
SV-10	Event-Driven Process Chain Model (EPC)
SV-11	Entity Relationship Model (ERM)

TV-1	*
TV-2	*

DoDAF views in ARIS

The following sections describe how the ARIS meta-model supports the DoDAF/C4ISR framework and how ARIS aligns at a high level to the CADM. Each DoDAF product definition is followed by a mapping to the ARIS meta-model and a graphical example of what the model looks like. Appendix A takes the analysis a step further and expands the data element and relationship tables found in the DoDAF to include ARIS mappings.

View Details – All Views

There are some overarching aspects of an architecture that relate to all three views. These overarching aspects are captured in the All-Views (AV) products. The AV products provide information pertinent to the entire architecture but do not represent a distinct view of the architecture. AV products set the scope and context of the architecture. The scope includes the subject area and timeframe for the architecture. The setting in which the architecture exists comprises the interrelated conditions that compose the context for the architecture. These conditions include doctrine; tactics, techniques, and procedures; relevant goals and vision statements; and concepts of operations, scenarios, and environmental conditions.

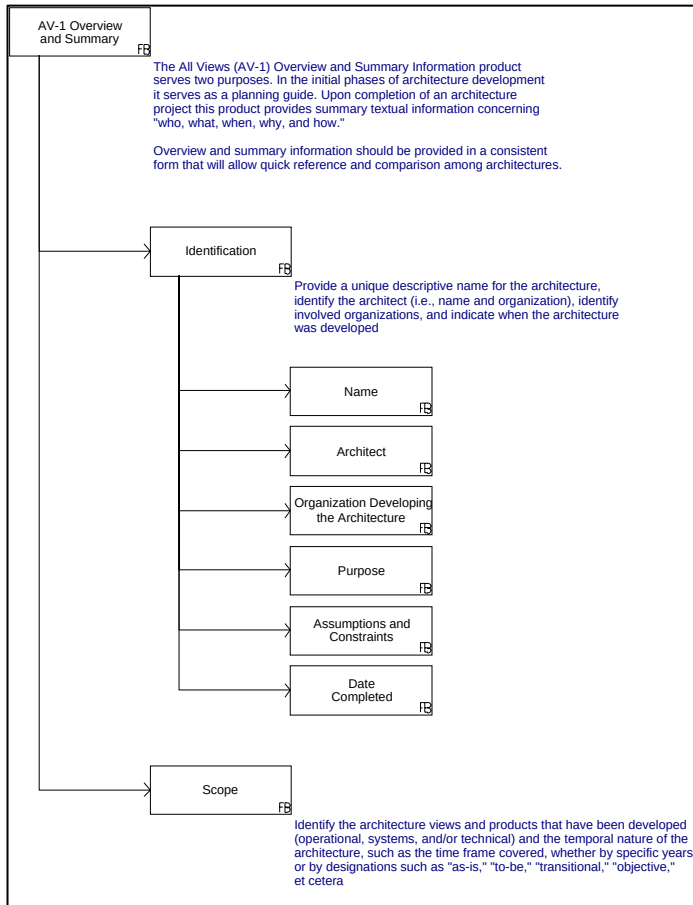
AV-1 OVERVIEW AND SUMMARY

DoDAF Brief Description:

The AV-1 product provides executive-level summary information in a consistent form that allows quick reference and comparison among architectures. The AV-1 product should contain sufficient information to enable a reader to select one architecture from among many to read in more detail. The AV-1 product includes assumptions, constraints, and limitations that may affect high-level decision processes involving this architecture.

The AV-1 explains the need for the architecture, what it should demonstrate, the types of analyses (e.g., Activity-Based Costing) that will be applied to it, who is expected to perform the analyses, what decisions are expected to be made on the basis of an analysis, who is expected to make those decisions, and what actions are expected to result. Also identifies the viewpoint from which the architecture is described (e.g., planner, owner, designer, or builder).

ARIS Example:



In ARIS, the AV-1 can be represented by a Technical Terms model and attribute information. The Technical Terms model provides an overview of architecture elements and their relationships.

Figure: AV-1 ARIS Model Example

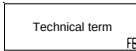
DoDAF Data Element	ARIS Data Element	ARIS Graphical Type
Topic	Technical Term	

Table: AV-1 Data Element Mapping

AV-2 DATA DICTIONARY

DoDAF Brief Description:

The Integrated Dictionary contains definitions of terms used in the given architecture. It consists of textual information in the form of a glossary, a repository of architecture data, their taxonomies, and their metadata (i.e., data about the data elements), including meta data for tailored products, associated with the architecture products developed.

ARIS Example:

The AV-2 can be composed of a Technical Terms model and report information. The Technical Terms models provides on overview of dictionary elements and their relationships. Architecture reports can provide a range of data based on architecture models. In a sense, the underlying data model *is* the majority of the dictionary.

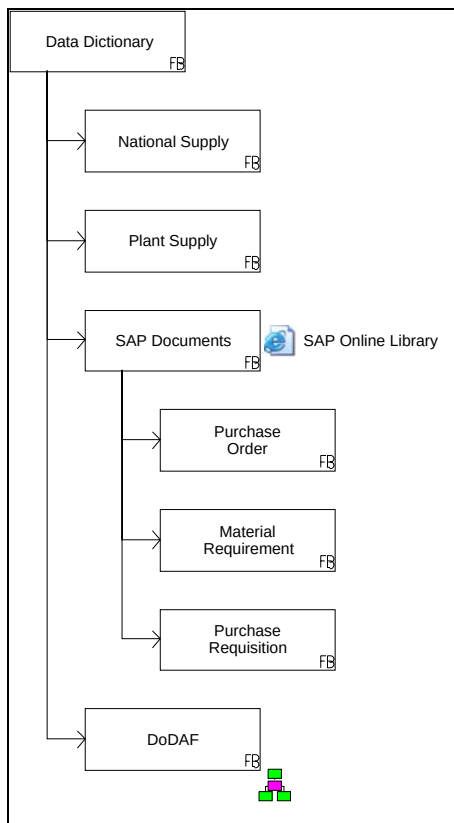


Figure: AV-2 ARIS Model Example

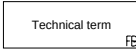
DoDAF Data Element	ARIS Data Element	ARIS Graphical Type
Topic	Technical Term	

Table: AV-2 Data Element Mapping

View Details - Operational Views

The operational architecture is a description of the tasks and activities, operational elements (nodes), and information flows required to successfully perform a mission. It contains descriptions (often graphical) of the operational elements, assigned tasks and activities, and information flows required to support the warfighter. It defines the types of information exchanged, the frequency of exchange, which tasks and activities are supported by the information exchanges, and the nature of information exchanges in detail sufficient to ascertain specific interoperability requirements.

OV-1 HIGH-LEVEL OPERATIONAL CONCEPT

DoDAF Brief Description:

The OV-1 consists of a graphical executive summary for a given architecture. The product identifies the mission/domain covered in the architecture and the viewpoint reflected in the architecture (e.g., joint commander, service commander, Principal Staff Assistant, support organization, intelligence unit). The OV-1 should convey, in simple terms, what the architecture is about and an idea of the players and operational capabilities involved.

ARIS Example:

The OV-1 shows a high level operational concept usually presented as elaborative graphic. It is possible to break down the graphical elements into basic components of organizations, activities, systems, and information and represent these elements in a more structured way using Value Added Chain Diagrams (VACD) or eEPC's. The advantage is that these basic elements can be linked to sub-architecture elements describing their relationships and behavior in detail. That allows a seamless drill down from high-level vision to its detail components.

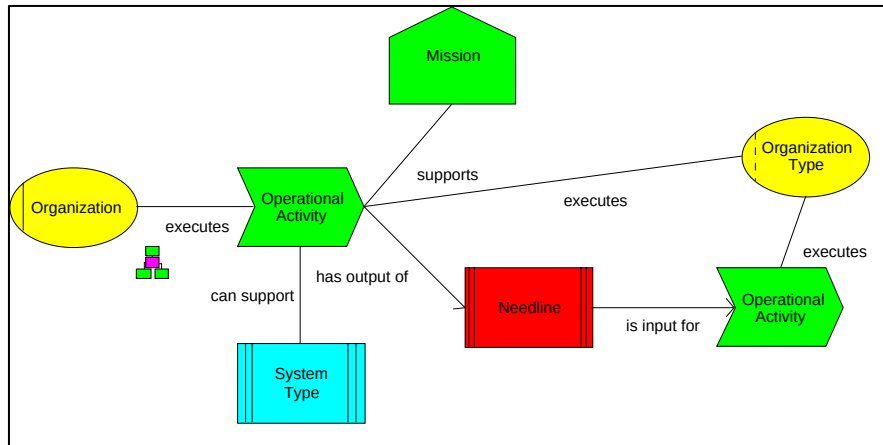
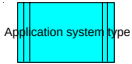


Figure: OV-1 ARIS Model Notation Example

DoDAF Data Element	ARIS Data Element	ARIS Graphical Type
Organization	Organization Unit	
Organization Type	Organization Unit Type	
System	Application System	
System Type	Application System Type	
Mission	Objective	

Operational Activity	Function	
Information	Cluster	

Table: OV-1 Data Element Mapping

OV-2 OPERATIONAL NODE CONNECTIVITY DIAGRAM

DoDAF Brief Description:

An operational architecture may be depicted at a high level through the use of a set of interacting operational nodes (or organizations) and the identification of types of information they exchange. OV-2 graphically depicts the operational nodes with needlines between those nodes that indicate a need to exchange information. The graphic includes internal operational nodes (internal to the architecture) as well as external nodes. An operational node is an element of the operational architecture that produces, consumes, or processes information. The notion of “node” will vary depending on the level of detail addressed by the architecture effort.

ARIS Example:

The OV-2 shows information flow between operational nodes from a high level view. Operational nodes are identified in ARIS as organizational elements and activities (functions). Needlines are not shown simply as direct line between nodes but rather as an information element that flows between nodes. The presence of such an element in ARIS has the advantage of drill down capabilities to a variety of model types (e.g., the needline can be decomposed into more detailed exchanges). The element can also own a large number of attributes and relationships.

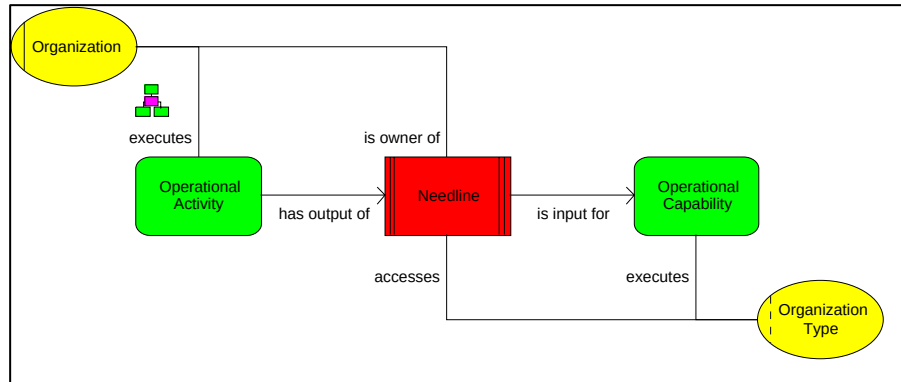


Figure: OV-2 ARIS Model Notation Example

DoDAF Data Element	ARIS Data Element	ARIS Graphical Type
<i>Organization</i>	Organization Unit	
Organization Type	Organization Unit Type	
Operational Activity	Function	
Information Flow/ Needline	Cluster	

Table: OV-2 Data Element Compliance

OV-3 OPERATIONAL INFORMATION EXCHANGE MATRIX

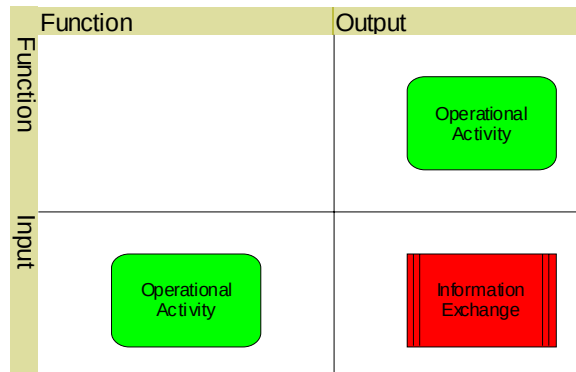
DoDAF Brief Description:

The Operational Information Exchange Matrix details Information exchanges that identify “who exchanges what information, with whom, why the information is necessary, and how the information exchange must occur.” [CJCSI 6212.01B] There is not a one-to-one mapping of OV-3 information exchanges to OV-2 needlines – rather, many individual information exchanges may be associated with one needline.

Information exchanges express the relationship across the three basic data elements of an OV (operational activities, operational nodes, and information flow) with a focus on the specific aspects of the information flow. Certain aspects of the information exchange can be crucial to the operational mission and should be tracked as attributes in the Information Exchange Matrix. The Operational Information Exchange Matrix contains details on the architecture’s “what” data elements, here called information exchanges. The matrix also relates these information exchanges to “who” (or operational nodes), and “how” (or business processes/operational activities of the OV-5), and “when” at the Owner levels of detail.

ARIS Example:

The OV-3 shows information exchanges between operational activities in a matrix form. ARIS provides the capability to show that matrix in a graphical format. It is also possible to automatically create a report table based on existing model information across views and export it to Office applications.



An IEDM model can also be used to represent the OV-3 to increase understanding of relationships between the information related elements: Information Exchange, Information Flow, Needline, and Interface. The model captures relations of those elements across views. Those relations are used to describe view-specific relationships and support enhanced reporting on the architecture.

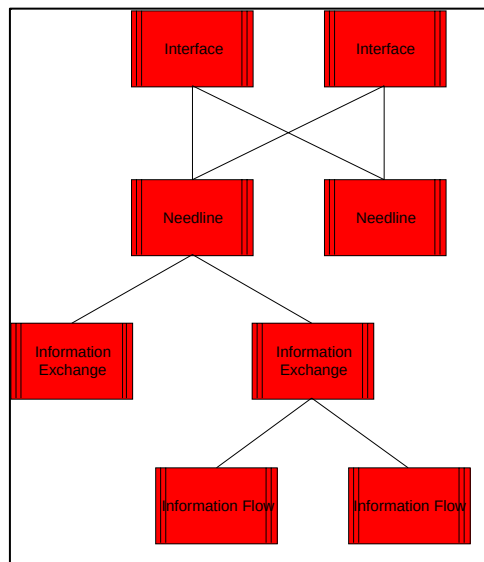


Figure: OV-3 ARIS Input/Output Matrix Notation Example

Figure: OV-3 ARIS IEDM Model Notation Example

DoDAF Data Element	ARIS Data Element	ARIS Graphical Type
Operational Activity	Function	
Information Exchange, Information Flow, Needline, Interface	Cluster	

Table: OV-3 Data Element Mapping

OV-4 ORGANIZATIONAL RELATIONSHIP CHART

DoDAF Brief Description:

The purpose of this product is to clarify the various relationships that can exist between organizations and sub-organizations within the architecture, and between internal and external organizations. The product illustrates the relationships among operational roles or organizations that are the key players in an architecture. These key players correspond to the operational nodes of an OV-2, which contains added detail on how the key players interact together in order to conduct their corresponding operational activities of OV-5.

ARIS Example:

The OV-4 shows relationships between organizations across all levels of the organization hierarchy. If OV-1 and OV-2 models are included in an ARIS supported architecture, a breakdown from organizational nodes down to the role/position level of the organization is possible.

The relationships between organizations and activities can be shown in OV-4 models or relationships can be mapped using existing model data from other views.

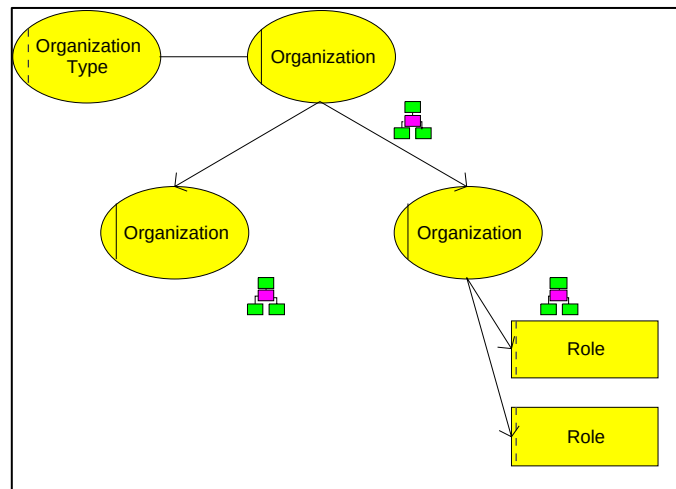


Figure: OV-4 ARIS Model Notation Example

DoDAF Data Element	ARIS Data Element	ARIS Graphical Type
<i>Organization</i>	Organization Unit	
Organization Type	Organization Unit Type	
Operational Activity	Function	
Role	Position Type	

Table: OV-4 Data Element Compliance

OV-5 OPERATIONAL ACTIVITY MODEL

DoDAF Brief Description:

The Operational Activity Model describes operational activities, the information exchanged between operational nodes (which conduct those activities), and the information exchanged with other activities outside the scope of the architecture (i.e., external exchanges) needed to effect the execution of a mission or goal as stated by the architecture purpose.

ARIS Example:

The OV-5 shows a hierarchy of operational activities and their relationships to organizations and information flow. The elements that can be captured in the OV-5 have close links to elements from other views and a number of relationships can be retrieved from those views. In ARIS, the OV-5 can be represented by a value added chain diagram (VACD). ARIS is a process-oriented tool and therefore is capable of showing sequential relations between operational activities, something not required in the OV-5.

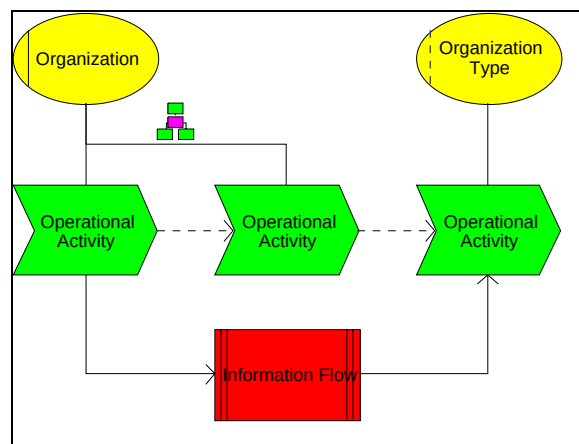


Figure: OV-5 ARIS VACD Model Notation Example

Another model, the ARIS function tree, allows the decomposition of operational activities and capabilities to the smallest functional entity..

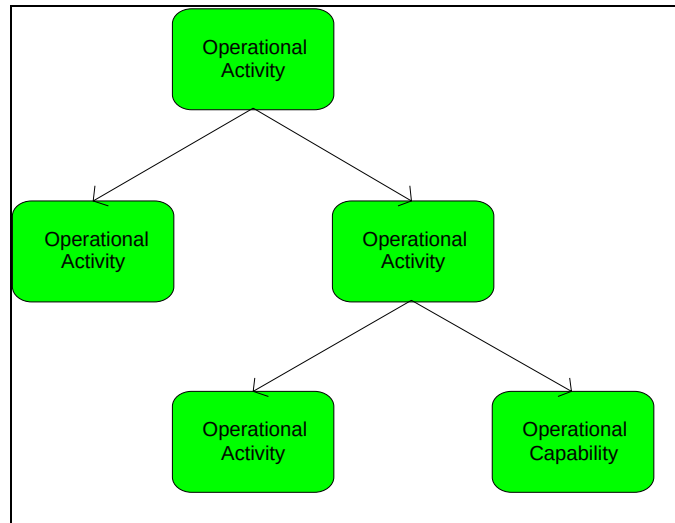


Figure: OV-5 ARIS Function Tree Model Notation Example

DoDAF Data Element	ARIS Data Element	ARIS Graphical Type
Operational Activity	Function	
Organization	Organization Unit	
Organization Type	Organization Unit Type	
Information Flow	Cluster	

Table: OV-5 Data Element Mapping

OV-6 OPERATIONAL RULE MODEL AND STATE TRANSITION

DoDAF Brief Description:

The DoDAF presents three components that make up the OV-6:

OV-6(a) Operational Rules Model: Operational or business rules are constraints on an enterprise, a mission, operation, business, or an architecture. While other OV products (e.g., OV-2, OV-5, OV-7) describe the structure of a business. At the mission level, the Operational Rules Model may consist of doctrine, guidance, rules of engagement, and so forth. At the operation level, rules may include such things as a military Operational Plan (OPLAN). At lower levels, the Operational Rules Model describes the rules under which the architecture or its nodes behave under specified conditions.

OV-6(b) Operational State Transition Description: The Operational State Transition Description is a graphical method of describing how an operational node or activity responds to various events by changing its state. The diagram basically represents the sets of events to which the architecture will respond (by taking an action to move to a new state), as a function of its current state. Each transition specifies an event and an action. The explicit time sequencing of activities in response to external and internal events is not fully expressed in OV-5. An Operational State Transition Description can be used to describe the explicit sequencing of the operational activities. Alternatively, an Operational State Transition Description can be used to reflect the explicit sequencing of actions internal to a single operational activity, or the sequencing of operational activities with respect to a specific operational node.

OV-6(c) The Operational Event-Trace Description: The Operational Event-Trace Description provides a time-ordered examination of the information exchanges between participating operational nodes as a result of a particular scenario. Each Event-Trace diagram should have an accompanying description that defines the particular scenario or situation. The Operational Event-Trace Description allows the tracing of actions in a scenario or critical sequence of events. With time proceeding from the top of the diagram to the bottom, a specific diagram lays out the sequence of information exchanges that occur between operational nodes for a given scenario.

ARIS Example:

The OV-6 describes process behavior by defining operational rules, states, transitions, and event characteristics. A common method of describing system behavior is the Unified Markup Language (UML). This method was the original method adapted by the DoDAF developers for process description. For those users wishing to use UML, ARIS supports all UML model types.

Nevertheless, the ARIS toolset provides another method for describing process behavior – the event driven Process Chain (EPC). The EPC has significant advantages over using UML diagrams to describe process behavior based on business process information. An EPC can combine all relevant objects in one single model and describe dynamic behavior as a process flow. In addition, the ARIS drill-down functionality allows a decomposition of the process across multiple levels of detail and enabling process abstraction as well as hierarchy. With an EPC it is possible to describe process rules, state transitions, and event triggers by combining several model elements. The description of the process flow can be expanded

by adding additional elements like roles, organizations, data objects, information flows, etc. The EPC is a superior method for a process oriented system design.

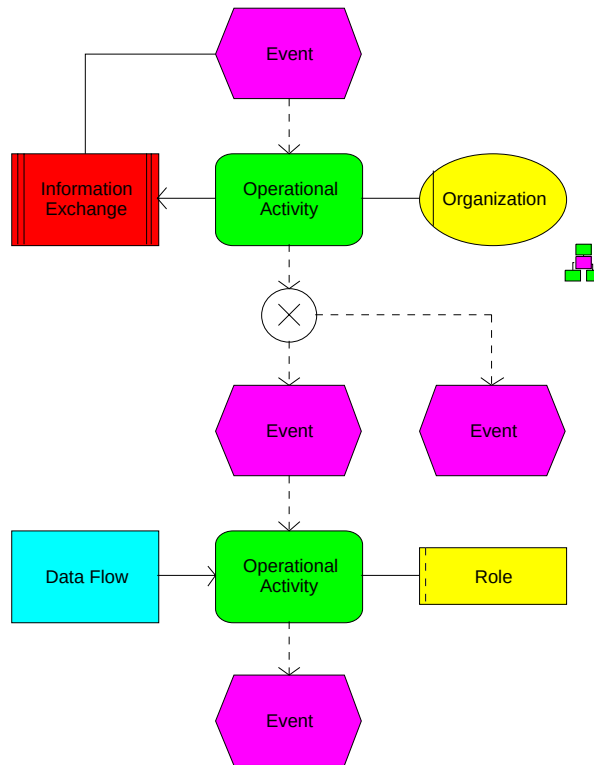


Figure: OV-6 ARIS EPC Model Notation Example

DoDAF Data Element	ARIS Data Element	ARIS Graphical Type
Operational Activity	Function	
Information Flow	Cluster	

<i>Event (State)</i>	Event	
<i>Organization</i>	Organization Unit	
Organization Type	Organization Unit Type	
Role	Person Type	
Condition	Rule	
Entity	Entity Type	

Table: OV-6 Data Element Mapping

OV-7 LOGICAL DATA MODEL

DoDAF Brief Description:

The Logical Data Model describes data requirements and structural business process rules of the architecture's operational view. It provides a definition of architecture data, their attributes or characteristics, and their interrelationships.

ARIS Example:

The Logical Data Model defines each kind of data associated with the mission or business as its own entity, with its associated attributes, and relationships. These data definitions correlate to the OV-2 needlines, the OV-3 information exchanges, and the OV-6c messages (information exchanges), as well as other products in the Architecture Framework. The OV-7 provides a data definition of architecture data elements. The data elements can be retrieved from various views of the DoDAF architecture. Data definitions are represented in an Entity Relationship Model (ERM) in ARIS.

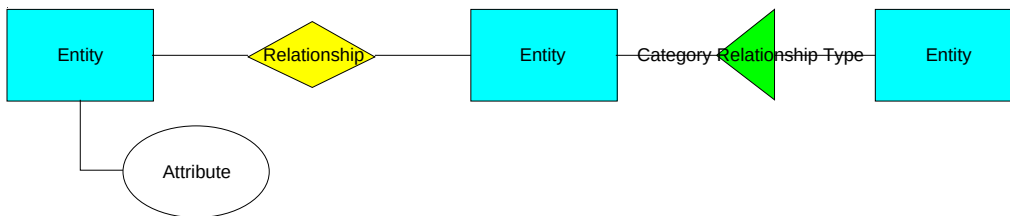


Figure: OV-7 ARIS ERM Model Notation Example

DoDAF Data Element	ARIS Data Element	ARIS Graphical Type
Entity	Entity Type	
Attribute	Attribute (ERM)	

Relationship	Relationship Type	
Category Relationship Type	Generalization	

Table: OV-7 Data Element Mapping

View Details - System Views

The systems views are a description, including graphics, of systems and interconnections providing for, or supporting, warfighting functions. For a domain, the systems architecture shows how systems link and interoperate, and may describe the internal construction and operations of particular systems within the architecture. For individual systems, the systems architecture includes the physical connection, location, and identification of key nodes (including materiel item nodes), circuits, networks, warfighting platforms, etc., and specifies system and component performance parameters (e.g., mean time between failure, maintainability, availability). The systems architecture aligns physical resources and their performance attributes to the operational architecture and its requirements per standards defined in the technical architecture.

SV-1 SYSTEM INTERFACE DESCRIPTION

DoDAF Brief Description:

The System Interface Description depicts systems nodes and the systems resident at those nodes to support organizational/role nodes of the Operational Node Connectivity Description (OV2). The System Interface Description also identifies the interfaces between Framework systems nodes, between Framework systems, and between the components of Framework systems.

ARIS Example:

The SV-1 represents the systems side of the organization at a node level in a similar manner to the way the OV-2 represents the organization from an operational perspective. It describes the interfaces between system nodes and allows for systems to be decomposed into subsystems. In ARIS, Application System Diagrams and Access Diagrams are used to describe interfaces and needlines between systems nodes, operational activities supported by systems, and a composition structure of systems.

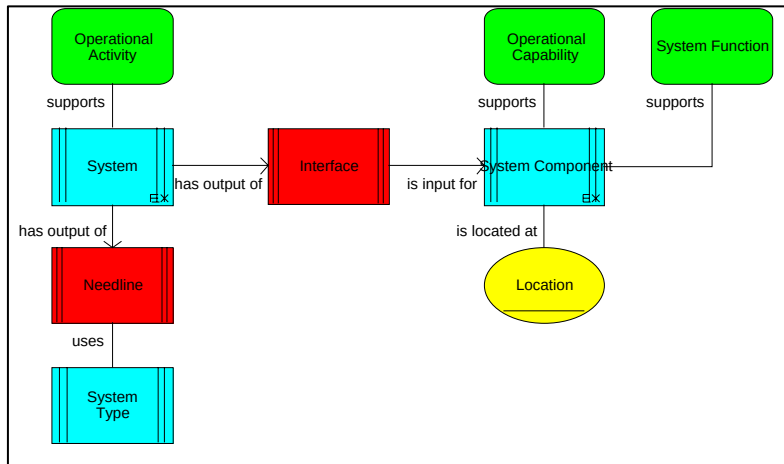


Figure: SV-1 ARIS Access Diagram Notation Example

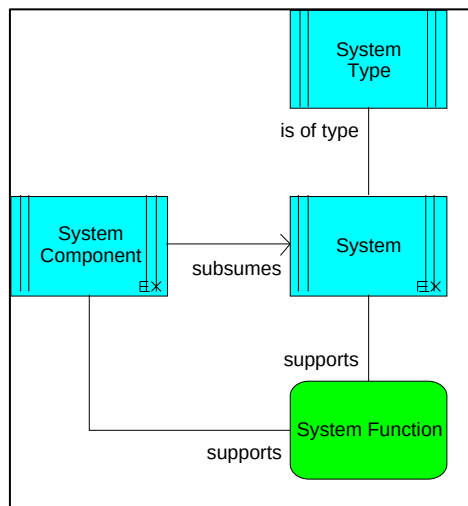


Figure: SV-1 ARIS Application System Diagram Notation Example

DoDAF Data Element	ARIS Data Element	ARIS Graphical Type
System	Application System	

System Type	Application System Type	
Operational Activity	Function	
Interface, Needline	Cluster	

Table: SV-1 Data Element Mapping

SV-2 SYSTEM COMMUNICATIONS DESCRIPTION

DoDAF Brief Description:

The Systems Communications Description depicts pertinent information about communications systems, communications links, and services (e.g., the kind of processing performed onboard a satellite, the locations of network switches or routers, the existence of amplifiers or repeaters in a particular communications link, or the location of cable “bulkheads” on both shores of an ocean). The Systems Communications Description documents the kinds of communications media that support the systems and their interfaces as described in SV-1.

ARIS Example:

The SV-2 describes the communications and networks that support the interfaces in the SV-1. ARIS offers a extensive functionality to describe networks graphically, including the Access and Network Diagrams. Elements from the SV-1 can be reused to map relationships in the SV-2 or new SV-2 Access Diagrams can be created.

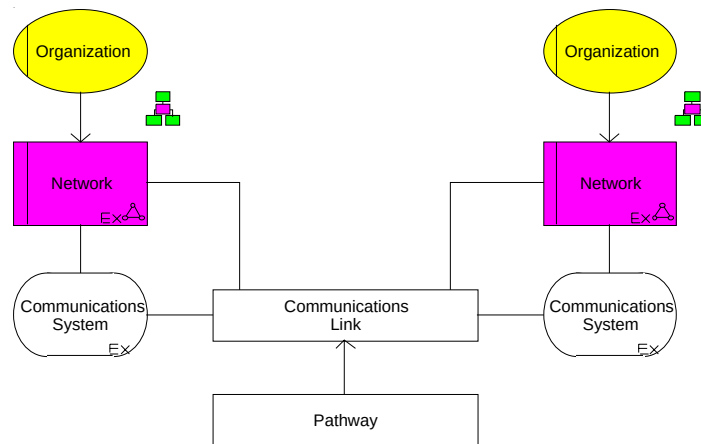


Figure: SV-2 ARIS Network Diagram Notation Example

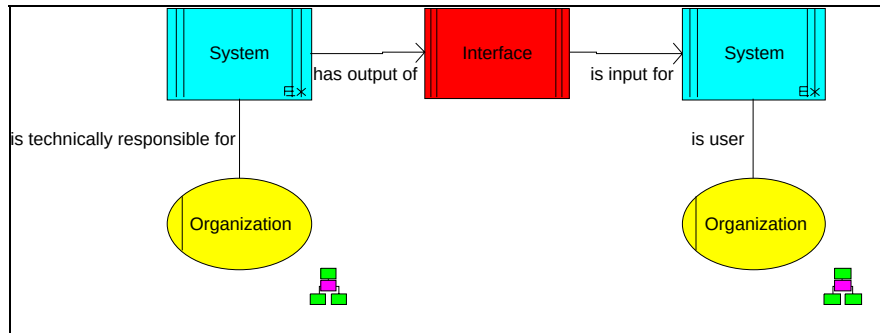


Figure: SV-2 ARIS Access Diagram Notation Example

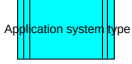
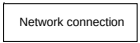
DoDAF Data Element	ARIS Data Element	ARIS Graphical Type
System	Application System	
System Type	Application System Type	
Organization	Organization Unit	
Communication System	Network	
Pathway	Network Connection	
Communications Link	Network Node	

Table: SV-2 Data Element Mapping

SV-3 SYSTEMS-SYSTEMS MATRIX

DoDAF Brief Description:

The Systems-Systems Matrix provides detail on the interface characteristics described in the Systems Interface Description for the architecture, arranged in matrix form. The Systems-Systems Matrix product allows a quick overview of all the interface characteristics presented in potentially several Systems Interface Description (SV-1) diagrams. The matrix form can support a rapid assessment of potential commonalities and redundancies (or, if fault-tolerance is desired, the lack of redundancies).

ARIS Example:

The SV-3 is a counterpart of an OV-3 in that it describes information flow between system nodes. The information can be retrieved from SV-1 and SV-2 models and modeled as a “graphical matrix” in an ARIS Access Diagram or generated in tabular format in the form of a report.

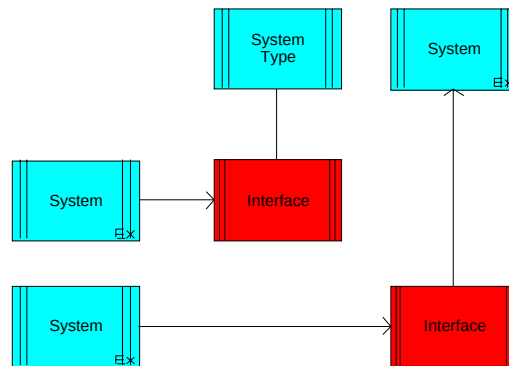
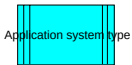


Figure: SV-3 ARIS Access Diagram Notation Example

DoDAF Data Element	ARIS Data Element	ARIS Graphical Type
System	Application System	
System Type	Application System Type	

Interface	Cluster	
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Table: SV-3 Data Element Compliance

SV-4 SYSTEMS FUNCTIONALITY DESCRIPTION

DoDAF Brief Description:

The Systems Functionality Description documents systems functional hierarchies and systems functions, and the data flows between them. This product focuses on describing the flow of data among system functions and is the systems view counterpart to the OV-5 Operational Activity Model of the OV.

ARIS Example:

A closer look at the ARIS eEPC, shown in the OV-6 example, reveals that the ARIS methodology allows the user to create a link between a function (operational activity) and a system. This enables us to ignore a distinction between an operational activity and a system function since all activities linked to a system are considered system functions. In other words, one and the same activity can be used on the operational side as well as the system side. The function does not have to be recreated to be used in a diagram representing a different perspective. Therefore, information from the OV-6 and OV-5 in ARIS can be used or mapped against SV-4 elements and the resulting information can be captured in an ARIS EPC.

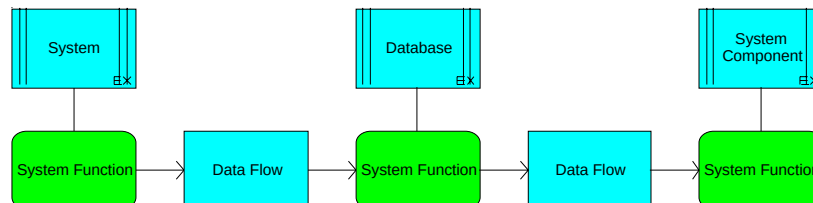


Figure: SV-4 ARIS EPC Notation Example

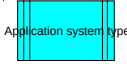
DoDAF Data Element	ARIS Data Element	ARIS Graphical Type
System	Application System	
System Type	Application System Type	
System Function	Function	
Entity	Entity Type	

Table: SV-4 Data Element Mapping

SV-5 OPERATIONAL ACTIVITY TO SYSTEMS FUNCTION TRACEABILITY DESCRIPTION

DoDAF Brief Description:

An Operational Activity to System Function Traceability Matrix correlates operational capability requirements that would *not* be satisfied if a specific system is *not* fielded to a specific unit in the architecture. It is a specification of the relationships between the set of operational activities applicable to an architecture and the set of system functions applicable to that architecture. This planner-level matrix allows decision makers to quickly identify stovepiped systems, redundant/duplicative systems, gaps in capability, and possible future investment strategies all in accordance with the time stamp given to the architecture.

ARIS Example:

The SV-5 is a mapping of system functions to operational activities. Like SV-4 information, SV-5 information is available in other models; e.g., inherited from existing relationships in the OV-5, OV-6, and SV-4. Therefore, a new model type is not required. ARIS reports can be designed to retrieve the information from the other views in order to provide the user with an SV-5.

SV-6 SYSTEMS DATA EXCHANGE MATRIX

DoDAF Brief Description:

The Systems Data Exchange Matrix specifies the characteristics and requirements of the data exchanged between systems. This product focuses on automated information and data exchanges that will be implemented in systems. The focus of the Systems Data Exchange Matrix is on how the data exchanges actually are implemented, in system-specific details covering such characteristics as specific protocols and data standard or media format.

ARIS Example:

The SV-6 describes the characteristics of the data flowing between systems. That information can be retrieved from a range of views via report tables. To improve understanding of relationships between various types of information flows used in different views in the DoDAF architecture, the user can create a diagram in ARIS (an IE Data Model) that represents those relationships.

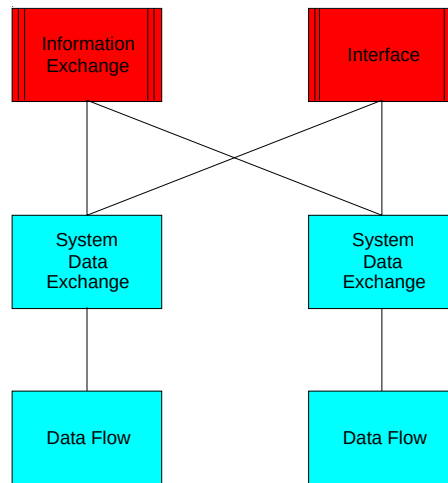


Figure: SV-6 ARIS IEDM Notation Example

DoDAF Data Element	ARIS Data Element	ARIS Graphical Type
System Data Exchange	Entity Type	
Data Flow	Entity Type	
Interface	Cluster	
Information Exchange	Cluster	

Table: SV-6 Data Element Mapping

SV-7 SYSTEMS PERFORMANCE PARAMETERS MATRIX

DoDAF Brief Description:

The Systems Performance Parameters Matrix product specifies the quantitative characteristics of systems and system components, their interfaces, and their functions. It specifies the current performance parameters of each system, system interface, or system function, and the expected or required performance parameters at specified times in the future.

Performance parameters include all technical performance characteristics of systems for which requirements can be developed and specification defined. The complete set of performance parameters may not be known at the early stages of architecture definition, so it should be expected that this product will be updated throughout the system's development, testing, and possibly even its deployment and operations life-cycle phases.

ARIS Example:

The SV-7 describes performance characteristics of systems and system functions. The relations can be describes in ARIS in a graphical format (in this case, a Key Performance Indicator Allocation Diagram) and mapped back to other architecture elements as well as to other model types. For example the performance information could be mapped back to a Balance Score Card (BSC) model.

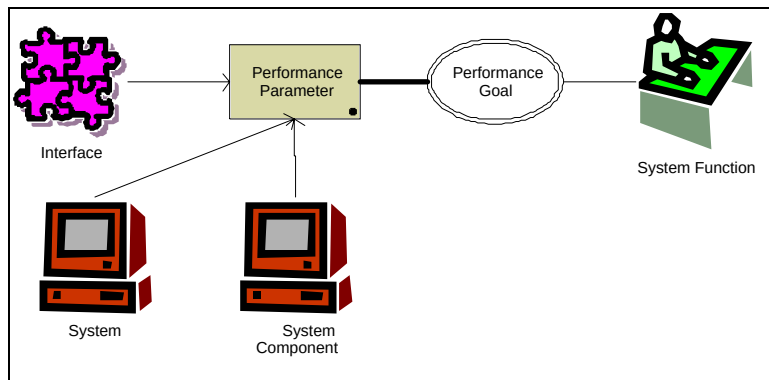


Figure: SV-7 ARIS Key Performance Indicator Allocation Diagram Notation Example

DoDAF Data Element	ARIS Data Element	ARIS Graphical Type
System, System Component	Application System	 Application system
Mission	Objective	

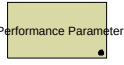
		
Performance Parameter	Key Performance Indicator	
Operational Activity	Function	
<i>Interface</i>	Cluster	

Table: SV-7 Data Element Mapping

SV-8: SYSTEMS EVOLUTION DESCRIPTION

DoDAF Brief Description:

Most architectures or systems are designed and deployed with some concepts of how the system, or the architecture in which the system is embedded, will be evolving over a lengthy period of time. The evolution plans are captured here in the Systems Evolution Description. Generally, the timeline milestones are critical for a successful understanding of the evolution timeline.

The Systems Evolution Description when linked together with other evolution products, provides a clear definition of how the architecture and its systems are expected to evolve over time. In this manner, the product can be used as an architecture evolution project plan.

ARIS Example:

Implementation and timeframe data can be captured as attributes of objects. This information would be available in reports. However, ARIS does not have a “timeline” model type with which to show this graphically.

SV-9: Systems Technology Forecast

DoDAF Brief Description:

The Systems Technology Forecast defines the underlying current and expected supporting technologies. It is not expected to include predictions of technologies as with a crystal ball. Expected supporting technologies are those that can be reasonably forecast, given the current state of technology and expected improvements. New technologies should be tied to specific time frames, which can correlate against the epochs used in the SV-8 milestones.

ARIS Example:

This information can be captured in the database, but is normally shown in text or tabular format. The information in the database can be seen through reports.

SV-10 SYSTEMS RULE MODEL

DoDAF Brief Description:

The DoDAF presents three components that make up the SV-10:

SV-10 (A): Systems Rules Model

Systems rules are constraints on an architecture, on a system(s) or system component(s), and/or on a system function(s). At the systems or systems components level, the Systems Rules Model describes the rules under which the architecture or its systems behave under specified conditions. The purpose of this product is to allow understanding of behavioral rules and constraints imposed on systems and system functions.

SV-10 (B): Systems State Transition Description

The Systems State Transition Description is a graphical method of describing a system (or system function) response to various events by changing its state. The diagram represents the sets of events to which the architecture will respond (by taking an action to move to a new state), as a function of its current state. Each transition specifies an event and an action. Basically, state transition diagrams can be unambiguously converted to structured textual rules that specify timing aspects of systems events and the responses to these events, with no loss of meaning. However, the graphical form of the state diagrams can often allow quick analysis of the completeness of the rule set, and detection of dead ends or missing conditions. These errors, if not detected early during the systems analysis phase, can often lead to serious behavioral errors in fielded systems, or to expensive correction efforts.

SV-10 (C): Systems Event-Trace Description

The Systems Event-Trace Description provides a time-ordered examination of the systems data exchanges between participating systems nodes as a result of a particular scenario. Each event trace diagram should have an accompanying description that defines the particular scenario or situation. The Systems Event-Trace Description in the systems view may reflect system-specific aspects or refinements

of critical sequences of events described in the operational view. The Systems Event-Trace Description allows the tracing of actions in a scenario or critical sequence of events. With time proceeding from the top of the diagram to the bottom, a specific diagram lays out the sequence of systems data exchanges that occur between system nodes for a given scenario. Different scenarios should be depicted by separate diagrams. The Systems Event-Trace Description can be used by itself or in conjunction with a Systems State Transition Description to describe dynamic behavior of system processes or system function threads.

ARIS Example:

The SV-10 describes system behavior by defining system rules, states, transitions, and event characteristics. A common method of describing system behavior is the Unified Markup Language (UML). ARIS supports all UML model types.

An SV-10 is the system view counterpart to the OV-6. As described in OV-6 section, the ARIS EPC method has many advantages for process oriented system design. Therefore, the same method is used for a SV-10.

As described in the OV-6 section, the fact that activities are interchangeable between the operational and the system view could make certain system models redundant. Nevertheless, from an architecture standpoint it is feasible to describe both views independently. On one side this could be done by different levels of detail, showing operational process on a high level and system processes on a more detailed level. On the other hand, one could distinguish between operational and system processes. That can be done if a system supports multiple processes and thus the system behavior can be described by combining parts of various operational processes into one system process.

DoDAF Data Element	ARIS Data Element	ARIS Graphical Type
System Function	Function	
Information	Cluster	

<i>Event (State)</i>	Event	
<i>Organization</i>	Organization Unit	
Organization Type	Organization Unit Type	
Role	Person Type	
Condition	Rule	  
Entity	Entity Type	

Table: SV-10 Data Element Mapping

SV-11 PHYSICAL SCHEMA

DoDAF Brief Description:

The Physical Schema product is one of the closest architecture products related to actual system design in the architecture framework. This product defines the structure of the various kinds of data that will be utilized by the systems in the architecture. The Physical Schema is an implementation-oriented data model that is used in the systems view to describe how the information requirements represented in the OV-7 Logical Data Model are actually implemented.

ARSI Example:

The SV-11 describes the relations of architecture data elements on implementation level. As in the OV-7, the ARIS ERM model is used for this view.

DoDAF Data Element	ARIS Data Element	ARIS Graphical Type
Entity	Entity Type	
Attribute	Attribute (ERM)	
Relationship	Relationship Type	
<i>Category Relationship Type</i>	Generalization	

Table: SV-11 Data Element Compliancy

View Details -Technical Views

The technical standards view provides the technical systems-implementation standards upon which engineering specifications are based, common building blocks are established, and product lines are developed. The TV includes two products, the Technical Standards Profile (TV-1) and the Technical Standards Forecast (TV-2) products.

TV-1 TECHNICAL STANDARDS PROFILE

DoDAF Brief Description:

This product collects the various legislative, policy, and other systems standards rules that constrain the choices that can be made in the design and implementation of an architecture. The Technical Standards Profile consists of the set of standards that govern systems implementation and operation of that architecture. The technical standards generally govern what hardware and software may be implemented and what systems data formats may be used. That is, only data formats, systems, system components, and communications protocols that are compliant with TV-1 standards may be used.

ARIS Example:

There have been no graphical models identified in ARIS to describe the technical standards profile. It is possible to use a Technical Terms Model to describe the structure of technical standards used in the architecture. Additionally, standards information can appear as attribute information for architecture elements. The information can be compiled afterwards into a report.

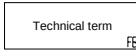
DoDAF Data Element	ARIS Data Element	ARIS Graphical Type
Standard Profile	Technical Term	

Table: TV-1 Data Element Compliance

TV-2 TECHNICAL STANDARDS FORECAST

DoDAF Brief Description:

The Technical Standards Forecast contains expected changes in technology-related standards and conventions which are documented in the TV-1 product. The forecast for evolutionary changes in the standards should be correlated against the time epochs as mentioned in the SV-8 and SV-9 products.

Technical Standards Forecast lists emerging or evolving technology standards relevant to the systems covered by the architecture. It contains predictions about the availability of emerging standards, and confidence factors for the predictions. It also contains matching predictions for market acceptance of each standard and an overall risk assessment associated with using the standard.

ARIS Example:

There have been no graphical models identified in ARIS to describe the technical standards forecast. It is possible to use a Technical Terms Model to describe the structure of a technical standards forecast used in the architecture. Additionally, standards information can appear as attribute information for architecture elements. The information can be compiled afterwards into a report.

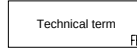
DoDAF Data Element	ARIS Data Element	ARIS Graphical Type
Standard Forecast	Technical Term	

Table: TV-2 Data Element Compliance

CADM Compliance

The Core Architecture Data Model (CADM) is the DoD standard logical data model for DoD architectures. It enables an efficient reuse and comparison of architectures and the required interoperability at the architecture data level.

The ARIS database metamodel is not build on a CADM compliant data model. This is true for other Business Process Management (BPM) Commercial Of The Shelf (COTS) tools. To circumvent this, it is possible to design an interface to exchange data between COTS tools and CADM-compliant databases (e.g., the DoD Architecture Repository System (DARS)⁷). In doing so, any business process modeling tool that can exchange structured data with a CADM-compliant database can be considered CADM-compliant.

In order to demonstrate that COTS architecture tools could communicate with a CADM-compliant database, OASD(NII) conducted the Architecture Data Interoperability Pilot (AIP) using DARS. IDS Scheer and EII participated in the pilot and showed that it is possible for ARIS and DARS to exchange architecture data for the OV-2, -4, and -5 products. ARIS is using standard XML export functionality mapped to All-DoD CADM specific XML tags to enable import into DARS. IDS Scheer and EII will continue to participate in the AIP to make all supported views interchangeable via CADM-XML.

⁷ DARS is an All-DoD CADM compliant repository that is intended to store, in a central location, legacy, draft, and approved architecture information developed by the Commands, Services and Agencies (C/S/As).

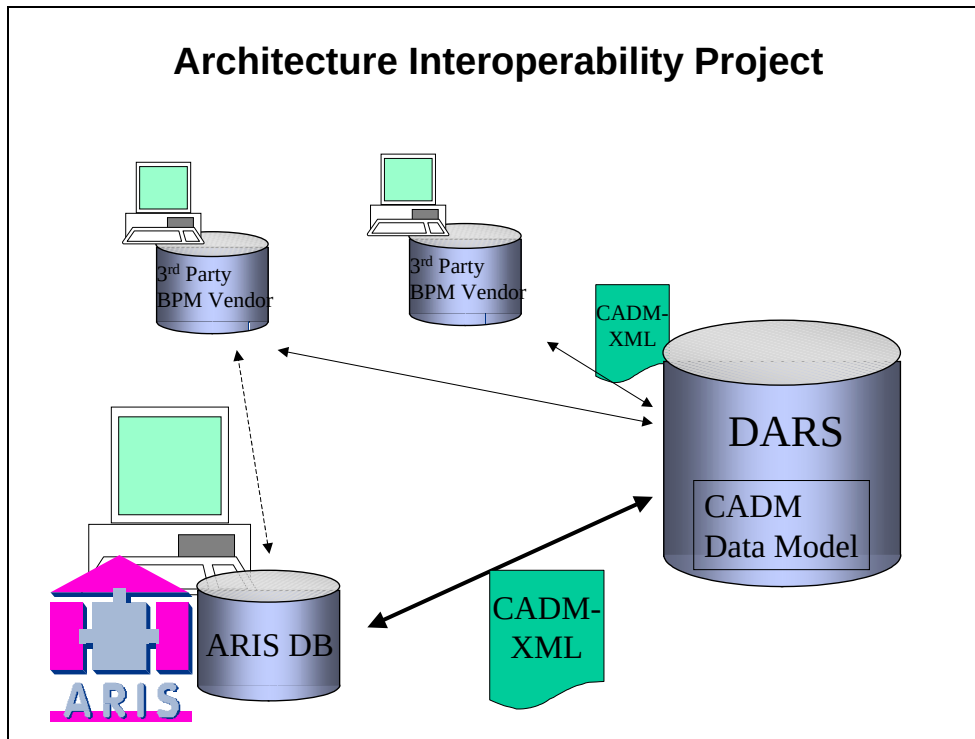


Figure: Architecture Interoperability Project

Building Architectures

“Enterprise Architecture Planning is the process of defining architectures for the use of information in support of the business and the plan for implementing those architectures.” Spewak, Steven H.

Building of an architecture is the process from defining requirements and selecting an architecture framework to defining the relevant methods and objects.

Approach

As the leading vendor of BPM tools, IDS Scheer has a well-defined method for – and vast experience in – building business architectures. The following steps describe the process of defining an architecture:

- Establish architecture requirements: Architecture requirements can be driven by a number of factors. The most important one is probably the objective and scope a specific initiative. For example, an SAP implementation should look at ASAP compliant architecture; a pure system design effort may consider an architecture with a UML focus. Besides that, architectures can be driven by organizations, standards, or legal requirements.

- Select a framework: The framework classifies business information elements and provides standards of representation. An selection of a framework should be driven by the following considerations:
 - Is the framework still state of the art or commonly known, e.g. IDEF vs. UML?
 - Does the framework identify relevant elements for my initiative?
 - Does the framework represent the elements in a desired context and format?
 - Is the framework flexible to incorporate organization specific requirements?
 - Does the framework meet standard, organizational, or legal requirements?
 - How easy is it to understand and teach the framework?
 - Is the framework supported by BPM tools?
- Select framework components that support requirements: A framework is usually very general to support a wide range of requirements. Not all elements and methods will be relevant for a specific initiative. This steps selects all elements and methods that are required to meet project goals.
- Select BPM tool that enables selected framework and components: All leading BPM tools provide functionality that supports a multitude of frameworks and methods. While the focus has to be on supporting the desired framework, the tool functionality plays an equally important role. The value of maintaining an architecture is determined on how it enables initiative objectives. Therefore the tool selection has to based on functionality that is tailored to specific initiatives like 6Sigma, ABC, BSC, or ERP implementations. It has to be based also on functionality that helps maintain, present, and retrieve architecture information, for example change management, web publishing, or reporting.
- Tailor framework based on detail requirements and tool functionality: The architecture-building process is not completed with the selection of a framework and its components. An architecture has to be structured in regards to architecture hierarchies, scope, documentation and other factors. This process transforms a generic framework into an organization- and initiative-specific architecture.

DoDAF Tailoring

The first two steps of the architecture development process above can be combined into one project preparation step. The major architecture requirements within the DoD organization are usually predefined based on compliance requirements defined by DoD and subordinate commands. A framework selection process is not necessary since all architectures within the DoD environment have to be DoDAF compliant.

The selection of framework components is driven in defense organizations by compliance requirements. The DoD Architecture Framework Volume 1.0 Volume II supplies a recommendation table of views that are applicable for specific types of projects.

The last step of the approach tailors the ARIS framework and functionality towards project specifics based on the DoDAF. The tailoring should include the following definitions:

- DoDAF views, objects, relations, and attributes that will be integrated into the architecture.
- ARIS model types, objects types, relationships, and attributes that map to the previous selection. (See Appendices A and B).
- Extension of the architecture to models and objects that are not required by DoDAF but are relevant for the project goals and objectives.
- Definition of architecture hierarchies and level of detail. Architectures are not flat pictures of information. They present a levels of detail build towards an breakdown of information from a management level to the required granularity.
- Definition of report tables that accompany the graphical representation of the information. The ARIS reporting functionality enables a user to predefine report outputs and layout and transfer architecture information into a predefined format. These reports can be used to generate all views that are not graphically supported in ARIS.

Examples

Conclusion

The DoD Architecture Framework, Version 1.0, defines a common approach for Department of Defense (DoD) architecture description development, presentation, and integration. It is intended to ensure that architecture descriptions can be compared and related across organizational boundaries, including Joint and multinational boundaries.

The DoD Architecture Framework is an open framework that allows adaptation to individual requirements as well as to existing frameworks and tools. It provides the necessary guidelines to create standardized and exchangeable architectures across DoD organizations and at the same time, enables openness and vendor neutrality.

ARIS allows the user to build a DoDAF- and All-DoD CADM-compliant architecture and enables the transfer of data to other CADM-compliant databases and 3rd party products. In addition, the additional capability and flexibility within ARIS allow users to customize the architecture to ensure it is a valuable tool in business process reengineering and application design.

Appendix A – Table: ARIS – DoDAF Mapping

Data Element Definitions for High-Level Operational Concept Graphic, OV-1

Data Elements	Attributes	ARIS Modeling	Example Values/Explanation
Graphical Box Types			Graphical Box Types
Asset Icon*		Can be represented as any node type object	
	Name*		The generic asset name that appears on the graphic (e.g., AWACS, fighter squadron, carrier battle group)
	Representation Type*		Type of asset represented by the icon e.g., platform, sensor, or weapon; organization; mission; or task (e.g., aircraft type; air organization; air assets; air mission or task)
	Description*		Textual description of the asset
	Generic Location		Location with respect to geographic configuration on graphic (e.g., open ocean, littoral, air, space, land, or specific land location or installation)
Target Area		Can be represented as any node type object	
	Identifier		Label on graphic or other assigned identifier
	Type		Type of target represented (e.g., land based installation, troops, satellite, aircraft, ships)
	Description		Text description of target importance or role
	Generic Location		Location with respect to geographic configuration on graphic
Graphical Arrow Types			
Line*			See OV-2 data element definition
	Identifier*		A generic line identifier
	Descriptive Name*		A generic name that indicates the kind of line represented (e.g., needline, systems interface, communications link, etc.)
	Description*		A description of the kind of line represented
	“From”		Name of the originating asset icon
	“To”		Name of the destination asset icon
Non-graphical Types			
Needline			See OV-2 data element definition

Data Elements	Attributes	ARIS Modeling	Example Values/Explanation
External Node			See OV-2 data element definition
Referenced Types			
Mission			See AV-1 data element definition
Operational Node			See OV-2 data element definition
Operational Capability or Activity			See OV-5 data element definition
Information Exchange			See OV-3 data element definition
Organization			See OV-4 data element definition
Systems Node			See SV-1/2 data element definition
System			See SV-1/2 data element definition
Relationships			
Asset Icon Represents an Operational Node, an Organization, a systems node, or a system			
	Asset Icon Name		Name of asset icon
	Operational Node, an Organization, a Systems Node, or a System Name		Name/identifier of operational node, an organization, a systems node, or a system represented by asset icon
Asset Icon is associated with Operational Capability or Activity			
	Asset Icon Name		Name of asset icon
	Target Area Name		Name/identifier of operational capability or activity associated with asset icon
Asset Icon is associated with Target Area			
	Asset Icon Name		Name of asset con
	Target Area Name		Name/identifier of target area associated with asset icon
Line represents needline, systems interface, communications link			See OV-2 data element definition
	Line Name		Name/identifier of line connecting asset icons

Data Elements	Attributes	ARIS Modeling	Example Values/Explanation
	Needline, systems interface, communications link Name		Name/identifier of line represented (e.g., needline name, systems interface name, communications link name, etc.)
Line Connects to Asset Icon			
	Asset Icon Name		Name of asset icon
	Line Name		Name/identifier of line connecting asset icon
Line Connects to Target Area			
	Target Area Name		Name of target area
	Line Name		Name/identifier of line connecting target area

Data Element Definitions for Operational Node Connectivity Description, OV-2

Data Elements	Attributes	ARIS Modeling	Example Values/Explanation
Graphical Box Types			
Operational Node*		Object: Function, Organizational Unit, Organizational Unit Type	
	Name*	Attrib: Name	Name or label of node box on diagram
	Description*	Attrib: Description/Definition	Text description of mission or role being performed by the node
	Level identifier	Attrib: Hierarchy Number, SAP Function Type	If using hierarchical decomposition of nodes: Identifier

Data Elements	Attributes	ARIS Modeling	Example Values/Explanation
			that corresponds to the node's place in the node hierarchy – should be unique
Graphical Arrow Types			
Needline*		Object: Cluster	
	Identifier*	Attrib: Identifier	Unique identifier of the needline (may be a number)
	Descriptive Name*	Attrib: Name	Descriptive name for the needline, usually associated with the type(s) of information associated with the needline
	Description*	Attrib: Description/Definition	Text description of needline
	“From” Operational Node*	Relationship: “has output of” between Function and Cluster Relationship: “is owner of” between Organizational Unit /Organizational Unit Type and Cluster (eEPC)	Name of node box that is the source of the node connector on the diagram
	“To” Operational Node*	Relationship: “is input for” between Function and Cluster Relationship: “accesses” between Organizational Unit /Organizational Unit Type and Cluster (eEPC)	Name of the node box that is the destination of the node connector on the diagram

Data Elements	Attributes	ARIS Modeling	Example Values/Explanation
External Needline*		See Needline	(if any). A needline which has one end not shown or a node outside the scope of the architecture
	Identifier*		Unique identifier of the needline (may be a number)
	Descriptive Name*		Descriptive name for the needline, usually associated with the type(s) of information associated with the needline
	Description*		Text description of needline
	“From” Node*		Name of node (operational node box or external node annotation) that is the source of the node connector on the diagram
	“To” Node*		Name of the node (operational node box or external node annotation) that is the destination of the node connector on the diagram
Hierarchy Chart Connector		Relationship: “Is superior to” connection between organizational units	

Data Elements	Attributes	ARIS Modeling	Example Values/Explanation
		is used in OV-4; “Is process-oriented superior” connection between functions (normally used in OV-5).	
	Parent Operational Node	Inherited by relationship	Name or level identifier of a node that has a decomposition
	Child Operational Node	Inherited by relationship	Name or level identifier of child (i.e., subordinate) node
Non-graphical Types			
External Node		See Operational Node	(if any) A node which is outside the scope of the architecture
	Name		Name of external node (i.e., label on diagram)
	Description		Textual description of the role performed by the external node
	Type		Logical (organizational) or Physical Node
Referenced Types			
Operational Capability		Object: Function	See OV-5 Definition Table
Operational Activity		Object: Function	See OV-5 Definition Table

Data Elements	Attributes	ARIS Modeling	Example Values/Explanation
Organization		Object: Organizational Unit	See OV-4 Definition Table
Organization Type		Object: Organizational Unit Type	See OV-4 Definition Table
Relationships			
Operational Node Is Decomposed Into Sub Operational Nodes		Normally shown in OV-4 and OV-5.	
	Operational Node	Inherited by relationship	Name or identifier of a decomposed operational node
	Sub Operational Node	Inherited by relationship	Name or identifier of a sub operational node
Operational Node Represents Organization		Attrib: DoDAF Discriminator OR relation in OV-4	
	Operational Node Name*	Inherited by relationship	Name/identifier of operational node representing the organization
	Organization Name*	Inherited by relationship	Name/identifier of organization represented by operational node
Operational Node Represents Organization Type		Attrib: DoDAF Discriminator OR relation in OV-4	
	External Node Name	Inherited by relationship	Name/identifier of operational node

Data Elements	Attributes	ARIS Modeling	Example Values/Explanation
			representing the organization type
	Organization Type Name	Inherited by relationship	Name/identifier of organization type represented by operational node
Operational Node is Associated with Operational Capability*		If Operational Capability is a node, then the occurrence of the Function object in the OV-2 model implies it is an operational node. If Organizations are nodes, then this is the “executes” connection between Organizational Unit and Function.	
	Operational Node Name*	Inherited by relationship	Name/identifier of operational node where the operational capability is performed
	Operational Capability Name*	Inherited by relationship	Name/identifier of operational capability associated with operational node
Operational Node Has Associated Operational Activity*		If Operational Activity is a node, then the occurrence of the Function object in the OV-2 model implies it is an operational node. If Organizations are nodes, then this is	

Data Elements	Attributes	ARIS Modeling	Example Values/Explanation
		the “executes” connection between Organizational Unit and Function.	
	Operational Node Name*	Inherited by relationship	Name/identifier of operational node where the operational activity is performed
	Operational Activity Name*	Inherited by relationship	Name/identifier of operational activity associated with operational node

Data Element Definitions for Operational Information Exchange Matrix, OV-3

Data Elements	Attributes	ARIS Mapping	Example Values/Explanations
Non-graphical Types			
Needline*		See OV-2 Definition Table	See OV-2 Definition Table
Information Exchange Identifier*		Object: Cluster	
	Identifier*	Attrib: Identifier	Identifier for the information exchange – usually based on the relevant Needline identifier; should be unique for the architecture
Information Element Identifier*		See Information Exchange	
	Identifier*	Attrib: Identifier	Identifier for the information element – usually based on the relevant information exchange identifier; should be unique for the architecture
	Name*	Attrib: Name	Name for the information element – indicative of the information

Data Elements	Attributes	ARIS Mapping	Example Values/Explanations
			content
	Content Description*	Attrib: Description/Definition	Description of the content of the information element
	Mission/Scenario UJTL or METL*		Joint Mission Area, cross-mission area domain, Universal Joint Task List activity, related specific scenario, or other task-related basis of the architecture
	Language	English	Identifier/name of codes of the natural language(s) involved in the information exchange. Relevant for multi-national operations
	Accuracy		Accuracy of data
	Transaction Type*	Attrib: Processing Code	Descriptive field that identifies the type of the information exchange. Valid values are: publish ⁸ , search ⁹ , subscribe ¹⁰ , collaborate ¹¹
	Triggering Event*	Relationship: "is state of" Relation between Event and Cluster in OV-6	Brief textual description of the event(s) that triggers the information exchange
	Interoperability Level		Level of Information Systems Interoperability (LISI) required, or other interoperability measure
	Periodicity*	Attrib: No of occurrences – avg.	How often the information exchange occurs; may be an average or a worst case estimate and may include conditions, e.g., wartime or peacetime
	Timeliness*		Required maximum allowable time of exchange from node to node (in seconds)

⁸ Publish: To make information products accessible.

⁹ Search: a) To ask for specific information. b) To provide services needed to identify and locate information products.

¹⁰ Subscribe: a) To select predefined information products. b) To place a user on a list for delivery of predefined information products.

¹¹ Collaborate: a) To interact with other entities on information products via multimedia. b) to assist users in establishing and terminating ad hoc or static collaboration via multimedia.

Data Elements	Attributes	ARIS Mapping	Example Values/Explanations
	IA - Availability Effort Code		The code that represents the relative level of effort required to be expended to ensure that the information can be accessed for use in a specific information-exchange-requirement-assurance
	IA - Confidentiality Type Code		The code that represents the kind of protection required for information to prevent unintended disclosure in a specific information-exchange-requirement-assurance
	IA - Dissemination Control Type Code		The code that represents the kind of restrictions on receivers of information for a specific information-exchange-requirement-assurance based on sensitivity of information
	IA - Information Criticality Code		The code that represents the seriousness of the benefit that the information exchange element provides to the objective of the action being taken for a specific information-exchange-requirement-assurance
	IA - Integrity Type Code		The code that represents the kind of requirements for checks that the content of the data element of a specific information-exchange-requirement-assurance has not been altered
	IA - Accountability Type Code		Security goal that ensures that responsibility for actions/events can be given to an organization willingly or by obligation
	Protection Duration Code		The code that represents how long the information must be safeguarded for a specific information-exchange-requirement-assurance
	Protection Elapsed-Time Quantity		The elapsed-time quantity that represents the length of time during which information assurance protections of the information is required for a specific information-exchange-

Data Elements	Attributes	ARIS Mapping	Example Values/Explanations
			requirement-assurance
	Protection Suspense Calendar Date		The calendar date on which the designated level of safeguarding discontinues for a specific information-exchange-requirement-assurance
	Classification*		Classification code for the information element
	Classification Caveat*		A set of restrictions on information of a specific classification. Supplements a security classification with information on access, dissemination, and other types of restrictions
Referenced Types			
Operational Activity*		Object: Function	See OV-5 Definition Table
	Producing Operational Activity Name	Attrib: Name	Name of the operational activity (at the originating node of the needline) that produces the information exchange
	Consuming Operational Activity Name	Attrib: Name	Name of the operational activity (at the receiving node of the needline) that consumes the information exchange
Event*		Relationship: Relation between Event and Function in OV-6	See OV-6c Definition Table
Relationships			
Needline Supports Information Exchange*		Relationship: "Consists of" connection between Clusters.	
	Needline Identifier*	Attrib: Identifier	Identifier of a needline
	Information Exchange	Attrib: Identifier	Identifier of an information exchange supported by the

Data Elements	Attributes	ARIS Mapping	Example Values/Explanations
	Identifier*		needline
Information Exchange Has Producing Operational Activity*		Relationship: “Has Output of” connection between Function and Cluster.	
	Operational Activity Name*	Attrib: Name	Name of the operational activity (at the originating node of the needline) that produces the information exchange
	Information Exchange Identifier*	Attrib: Identifier	Identifier of the information exchange produced
Information Exchange Has Consuming Operational Activity*		Relationship: “Has Input for” connection between Function and Cluster.	
	Operational Activity Name*	Attrib: Name	Name of the operational activity (at the receiving node of the needline) that consumes the information exchange
	Information Exchange Identifier*	Attrib: Identifier	Identifier of the information exchange consumed
Information Exchange Has Triggering Event*		Relationship: “is state of” Relation between Event and Cluster in OV-6 OR Derived from eEPC logic in OV-6	
	Event Name*	Attrib: Name	Name of the event that triggers the information exchange
	Information Exchange Identifier*	Attrib: Identifier	Identifier of the information exchange triggered

Data Element Definitions for Organizational Relationships Chart, OV-4

Data Elements	Attributes	ARIS Mapping	Example Values/Explanation
Graphical Box Types			
Organization*		Object: Organizational Unit	
	Name*	Attrib: Name	Name of organization that appears on the graphic
	Description*	Attrib: Description/Definition	Text description of the organization's purpose, including the spelling out of all acronyms
	(Military) Service	Attrib: Short Description	Army, Navy, Air Force, Marine Corps, Joint
	Code/Symbol		Service office code or symbol
	Role/Responsibility		Text description of the role played by the described organization
Human Role		Object: Person Type	
	Name	Attrib: Name	Name of a human role (personnel position) in an organization
	Description	Attrib: Description/Definition	Text description of the human role
	Role/Responsibility		Text description of the responsibility the human role performs
Graphical Arrow Types			
Organizational		Represented by	

Data Elements	Attributes	ARIS Mapping	Example Values/Explanation
Relationship*		connection types between organizational objects.	
	Name/Label*	Attrib: Connection role	Relationship label used on graphic
	Description*	Attrib: Description/Definition	Textual description of relationship
	Relationship Type*	Attrib: Type	The type of the relationship documented (e.g., Direct/Command, Indirect, Situation Dependent; Coordination; Backup)
	Organization Name 1*	Inherited by relationship	Name of source organization for relationship
	Organization Name 2*	Inherited by relationship	Name of destination organization for relationship
Non-graphical Types			
Organization Type		Object: Organizational Unit Type	
	Name	Attrib: Name	Name of organization type
	Description	Attrib: Description/Definition	Text description of the organization type
Referenced Types			
Operational Node		Object: Organizational Unit	See OV-2 Definition Table

Data Elements	Attributes	ARIS Mapping	Example Values/Explanation
		Type	
Operational Activity		Object: Function	See OV-5 Definition Table
Relationships			
Organization Owns Operational Node		See OV-2 description.	
	Organization Name		Name of an organization or organization type
	Operational Node Name		Name of an operational node owned/operated by the organization
Organization Performs Operational Activity		Relationship: “executes” relation between Organization Unit/Organization Unit Type and Function	
	Organization Name	Inherited by relationship	Name of an organization or organization type
	Operational Activity Name	Inherited by relationship	Name of an operational activity that is performed (in whole or in part) by personnel or resources from the organization
Organization has Human Role		Relationship: “Belongs to” connection between Organizational Unit and Person Type	
	Organization Name	Inherited by	Name of an

Data Elements	Attributes	ARIS Mapping	Example Values/Explanation
		relationship	organization or organization type
	Human Role Name	Inherited by relationship	Name of a human role
Human Role Performs Operational Activity		Relationship: “executes” relation between Person Type and Function on OV-6	
	Human Role Name	Inherited by relationship	Name of a human role
	Operational Activity Name	Inherited by relationship	Name of an operational activity that is performed by human role

Data Element Definitions for Operational Activity Model, OV-5

Data Elements	Attributes	ARIS Mapping	Example Values/Explanation
Graphical Box Types			
Operational Capability*		Object: Function	
	Name*	Attrib: Name	Name of operational capability
	Description*	Attrib: Description/Definition	Description of the capability
	Line of Business*	Attrib: Short Description	One of the lines of business in OMB’s BRM
Operational Activity*		Object: Function	
	Name*	Attrib: Name	Name of activity
	Level	Attrib: Hierarchy	If using hierarchical

Data Elements	Attributes	ARIS Mapping	Example Values/Explanation
	identifier	Number OR SAP Function Type	decomposition of activities: Identifier that corresponds to the activity's place in the activity hierarchy – should be unique
	Description*	Attrib: Description/Definition	Description of the activity
	References	Attrib: Link + Title	Any policy or doctrine references that provide further explanation of the activity
	Operational Activity Cost	Attrib: Avg. total cost	Cost for activity derived from or used in activity based costing analysis
Graphical Arrow Types			
Hierarchy Chart Connector		Relationship: "Process-oriented superior" relationship between Functions	
	Parent Operational Activity	Inherited by relationship	Name or level identifier of an activity that has a decomposition
	Child Operational Activity	Inherited by relationship	Name or level identifier of child (i.e., subordinate) activity
Flow Connector*		Relationship: "Is predecessor" relationship between functions	(For process oriented models)

Data Elements	Attributes	ARIS Mapping	Example Values/Explanation
		OR inherited by relations between Function – Cluster – Function OR inherited by relations in OV-6	
	Name*	Attrib: Connection role	Label of connector on graphic
	Type		If using IDEF0: One of Input, Output, Control, Mechanism
For subtype Input/Output		N/A	
	Source*		Name of source activity or “External”
	Destination*		Name of destination activity or “External”
	Operational Activity I/O Information Name*		Name of the information associated with the flow
For subtype Control		N/A	(for IDEF0 models, if any)
	Source		Name of source activity or “External”
	Destination		Name of destination activity
	Control Description		Description of the control information or name of the operational activity I/O Information if the control is an

Data Elements	Attributes	ARIS Mapping	Example Values/Explanation
			output from an activity
For subtype Mechanism		N/A	(for IDEF0 models, if any)
	Source		Name of source activity, if relevant, otherwise "N/A"
	Destination		Name of destination activity
	Resource type		Type of resource represented: role or system
Non-graphical Types			
Model*		ARIS Model	
	Name*	Attrib: Name	Name /identifier of operational activity model
	Type*	Attrib: Type	Operational activity hierarchy, IDEF0-style model, object-oriented, or other type of model
	Purpose*	Attrib: Description/Definition	Purpose of model
	Viewpoint*		Viewpoint of model
Diagram*		ARIS Model is equal to a diagram	
	Title*	Attrib: Name	Title of diagram/graphic
	Diagram Number*		Level number of diagram (for leveled families of diagrams)
Facing Page		N/A	

Data Elements	Attributes	ARIS Mapping	Example Values/Explanation
Text			
	Identifier		Identifier/title of a page of text
	Text		Text description of the content of a graphic/diagram and its constituent parts
Operational Activity I/O Information*		Relationship: “Has output of” and “is input for” connections between Function and Cluster.	
	Name*	See: Information Exchange	Name of input or output information (or product) for an activity
	Description*	See: Information Exchange	Description of the information or product
Referenced Types			
Operational Node*		Object: Organizational Unit, Organizational Unit Type	See OV-2 Definition Table
Information Exchange*		Object: Cluster	See OV-3 Definition Table
Relationships			
Diagram Belongs To Model		N/A	
	Diagram Title		Title of a diagram
	Model Name		Name of the model to which the diagram belongs

Data Elements	Attributes	ARIS Mapping	Example Values/Explanation
Facing Page Text References Diagram		Metadata for the model itself, not model object contents	
	Facing Page Text Identifier		Identifier/title for a page of text
	Diagram Title		Title of the diagram which the text describes
Operational Activity Is Contained in Diagram		Occurrences in object properties	
	Operational Activity Name		Name/identifier of an activity
	Diagram Title		Title of the diagram on which the activity occurs.
Flow Connector Is Contained in Diagram		Occurrences in object properties	
	Flow Connector Name		Label of connector
	Diagram Title		Title of diagram on which the connector appears
Operational Activity Has Input		Relationship “Is input for” connection between Cluster and Function	
	Operational Activity	Inherited by relationship	Name or level identifier of an activity

Data Elements	Attributes	ARIS Mapping	Example Values/Explanation
	Operational Activity I/O Information Name	Inherited by relationship	Name or label of information or product that is an input to the activity
Operational Capability Corresponds To a Level Zero Operational Activity in an Operational Activity Model		N/A (IDEF0), relation can be retrieved in OV-4	
	Operational Capability		Name or identifier of an operational capability
	Operational Activity		Name or identifier of an operational activity
Operational Activity Is Decomposed Into Sub Operational Activity		Relationship: "Process Oriented Superior" relationship between Functions in OV-4	
	Operational Activity	Inherited by relationship	Name or identifier of a decomposed operational activity
	Sub Operational Activity	Inherited by relationship	Name or identifier of a sub operational activity
Operational Activity Has Output		Relationship: "Has output of" connection between Function and Cluster	
	Operational	Inherited by	Name or level identifier of an

Data Elements	Attributes	ARIS Mapping	Example Values/Explanation
	Activity	relationship	activity
	Operational Activity I/O Information Name	Inherited by relationship	Name or label of information or product that is an output from the activity
Operational Activity I/O Information Is Associated With An Information Exchange*		Relationship: “Has output of” and “is input for” connections between Function and Cluster. See OV-3 Definition Table.	(Information produced or consumed by an activity may be associated with multiple information exchanges.)
	Operational Activity I/O Information Name*	Inherited by relationship	Name or label of information that is an input or output of an activity
	Information Exchange Identifier*	Inherited by relationship	Identifier for an associated information exchange (from the Operational Information Exchange Matrix)
Operational Activity Is Performed At Node*		Relationship: “Executes” relationship between Organizational Unit/Organizational Unit Type and Function.	(Should be the inverse of the relationship operational node has associated operational activity)
	Operational Activity Name*	Inherited by relationship	Name/identifier of an activity
	Operational Node Name*	Inherited by relationship	Name/identifier of the operational node where that

Data Elements	Attributes	ARIS Mapping	Example Values/Explanation
			activity is performed.
Flow Connector Corresponds To Flow Connector		Relationship: "Consists of" connection between Clusters. See OV-3 description.	
	Flow Connector Name	Inherited by relationship	Name of flow connector on boundary of child diagram
	Flow Connector Name	Inherited by relationship	Name of the corresponding flow connector on parent diagram

Data Element Definitions for Operational Rules Model, OV-6a

Data Elements	Attributes	ARIS Mapping	Example Values/Explanation
Non-graphical Types			
Rule*		Inherent in eEPC logic.	
	Name*		Rule name/identifier
	Type*		One of: Structural Assertion, Action Assertion, Derivation
	Description*		Textual discussion of assertion or derivation
	Text*		Text of assertion or derivation in selected formal language
Formal Language*		N/A	

Data Elements	Attributes	ARIS Mapping	Example Values/Explanation
	Name*		Name of the formal language selected to record the rules
	Source*		Reference to the textbook or article where the selected formal language is described.
Referenced Types			
Operational Activity*			See OV-5 Definition Table
Entity*			See OV-7 Definition Table
Relationships			
Rule Applies to Operational Activity*		Inherent in eEPC logic.	(if any rules are related to activities)
	Rule Name*		Name of action assertion or derivation rule
	Operational Activity Name*		Name of activity to which the rule applies
Rule Applies to Entity*		Inherent within the Input/Output relationships between Functions and Clusters and Functions.	(if any rules are related to entities)
	Rule Name*		Name of structural assertion rule
	Entity Name*		Name of entity to which the rule applies

Data Elements	Attributes	ARIS Mapping	Example Values/Explanation
Graphical Box Types			
State*		Object: Event	
	Name*	Attrib: Name	State name
	Description*	Attrib: Description/Definition	Textual description as necessary
	Type*	Attrib: Processing Code	One of: Simple, Nesting, Concurrent Superstate
	For Concurrent Superstates	N/A	
	Number of Partitions	Inherited by relationships or sub-models	Number of contained state charts
Graphical Arrow Types			
Transition*		Event-Function-Event construct	
	Label*		Identifier or event that triggers the transition
	Description*		Textual description of transition
	Type*		One of: Simple, Splitting, Synchronizing
	For Simple Transitions*		
	Source State Name*		Name of state where transition begins
	Target State Name*		Name of state where transition

Data Elements	Attributes	ARIS Mapping	Example Values/Explanation
			ends
	For Splitting Transitions		
	Number of Target States		Number of states where transition ends
	For Synchronizing Transitions		
	Number of Source States		Number of state where transition begins
	Target State Name		Name of state where transition ends
Non-graphical Types			
State Chart*		Model Type: eEPC, (UML State Chart)	
	Name*	Attrib: Name	Name/identifier of state chart
	Description*	Attrib: Description/Definition	Textual description of what the state chart represents
	Start State Name*	Inherent in eEPC logic.	Name of start state for state chart
Action*		Function	
	Name*	Attrib: Name	Name/identifier of an action that takes place while the system is in a given state or during a state transition
	Description*	Attrib: Description/Definition	Pseudo-English or code for activity

Data Elements	Attributes	ARIS Mapping	Example Values/Explanation
			function
Event*		Event	
	Name*	Attrib: Name	Name of event
	Description*	Attrib: Description/Definition	Textual description of the event
Event Qualifier Attribute			
	Name		Name of attribute associated with an event
	Definition		Textual definition of attribute
Event Qualifier Action*			
	Name*		Name/identifier of action associated with an event
	Description*		Pseudo-English or code for action function
Event Qualifier Guard			
	Name		Name/identifier for a Boolean expression that must be true for the associated transition to trigger
	Definition		Expression that defines the guard
Event Qualifier			

Data Elements	Attributes	ARIS Mapping	Example Values/Explanation
Export Event			
	Name		Name of an event that will be exported beyond the scope of the generating state chart
	Description		Textual description of the event represented
Referenced Types			
Operational Activity*		Object: Function	See OV-5 Definition Table
Rule			See OV-6a Definition Table
Relationships			
State Chart Has Terminal State*		Final Event in eEPC or Process Interface to connect to next process.	
	State Chart Name*		Name/identifier of a state chart
	State Name*		Name of a terminal state for that state chart
Event Triggers Transition*		Event-Function Construct	
	Transition Name*		Name/identifier of a transition
	Event Name*		Name of the event that triggers the transition
Transition Has Event			

Data Elements	Attributes	ARIS Mapping	Example Values/Explanation
Qualifier Attribute			
	Transition Name		Name/identifier for a transition
	Event Qualifier Attribute Name		Name of attribute that characterizes the transition
Transition Has Event Qualifier Action*			
	Transition Name*		Name/identifier for a transition
	Event Qualifier Action Name*		Name of activity performed as a result of triggering the transition
Transition Has Event Qualifier Guard*			
	Transition Name*		Name/identifier for a transition
	Event Qualifier Guard Name*		Name of associated expression that must be true before transition can be triggered
Transition Has Event Qualifier Export Event			
	Transition Name		Name/identifier for a transition
	Event		Name of event that

Data Elements	Attributes	ARIS Mapping	Example Values/Explanation
	Qualifier Export Event Name		will be exported beyond the scope of the containing state chart as a result of triggering the transition
State Has Associated Action*		Event-Function Construct	
	State Name*		Name of a state
	Action Name*		Name of the action performed while the activity is in a given state
Transition Has Associated Action*		Event-Function Construct	
	State Name*		Name of a state
	Action Name*		Name of the action performed during a transition
Event Qualifier Guard Is Associated With Rule			
	Event Qualifier Guard Name		Name of associated expression that must be true before transition can be triggered
	Rule Name		Name of associated rule (of type action assertion)

Data Elements	Attributes	ARIS Mapping	Example Values/Explanation
Splitting Transition Has Ending State		eEPC logic and rules	
	Transition Name		Name/identifier of a splitting transition
	State Name		Name of one of the target states of the splitting transition
Synchronizing Transition Has Starting State		eEPC logic and rules	
	Transition Name		Name/identifier of a synchronizing transition
	State Name		Name of one of the source states for the synchronizing transition
Nesting State Has Contained State Chart			
	State Name		Name of nesting state
	State Chart Name		Name of the state chart that decomposes the nesting state
Concurrent Superstate Has Partition State Chart			
	State Name		Name of concurrent super

Data Elements	Attributes	ARIS Mapping	Example Values/Explanation
			state
	State Chart Name		Name of the state chart in one of the partitions
Splitting Transition Has Matching Synchronizing Transition		eEPC logic and rules	
	Splitting Start State Name		Name of a state that is the source for a splitting transition
	Synchronizing End State Name		Name of the target state where a synchronizing transition brings together the separate threads of control started by the corresponding splitting transition. Splitting and synchronizing transitions must come in corresponding pairs; each pair makes up a complex transition
Action Is Related To Operational Activity*		Connections between Functions. See OV-2 and OV-5 descriptions.	
	Action Name*		Name/identifier of an action
	Operational		Name/identifier of

Data Elements	Attributes	ARIS Mapping	Example Values/Explanation
	Activity Name*		an activity from the Operational Activity Model
	Relationship Description*		Text description of the relationship (e.g., action is same as activity, action is contained in activity, action contains activity)
Event Qualifier Action Is Related To Operational Activity*			
	Event Qualifier Action Name*		Name of an event qualifier action
	Operational Activity Name*		Name/identifier of an activity from the Operational Activity Model
	Relationship Description*		Text description of the relationship (e.g., event qualifier action is same as activity, event qualifier action is contained in activity, event qualifier action contains activity)

Data Element Definitions for Operational Event-Trace Description, OV-6c

Data Elements	Attributes	ARIS Mapping	Example Values/Explanation
Graphical Box Types			
Node Lifeline*		See OV-2 description.	
	Node Lifeline Name*		Name of the node lifeline
	Description*		Text description of any assumptions or scope constraints on the lifeline
Graphical Arrow Types			
Message*		Input/Output connections in the model	
	Name*		Message label or name of message
	Description*		Textual description of message
	Originating Node Lifeline Name*		Name of node lifeline where Message begins
	Terminating Node Lifeline Name*		Name of node lifeline where Message ends
Non-graphical Types			
Message Time*			
	Identifier*		Identifier for the time when an message stops or starts

Data Elements	Attributes	ARIS Mapping	Example Values/Explanation
	Position On Lifeline*		Relative position of message on lifeline (e.g., top-most or first, second, etc.)
	Formula		Algebraic formula for calculating the time of message occurrence (i.e., starting or stopping of the message) relative to beginning of node lifeline
Referenced Types			
Operational Node*			See OV-2 Definition Table
Information Exchange*			See OV-3 Definition Table
Action*			See OV-6b Definition Table
Event*			See OV-6b Definition Table
Relationships			
Node Lifeline Represents An Operational Node*		See OV-2 description.	
	Node Lifeline Name*		Name of the node lifeline
	Operational Node Name*		Name of the operational node represented by the node lifeline
Message Starts At Time*			
	Message Name		Label of the message on the graphic
	Starting		Identifier of the time at

Data Elements	Attributes	ARIS Mapping	Example Values/Explanation
	Message Time Identifier*		which the message occurs or starts; gives the relative position of the message on its starting lifeline; may be identical to the ending time
Message Ends At Time*			
	Message Name*		Label of the Message
	Ending Message Time Identifier*		Identifier of the time at which the message ends; gives the relative position of the message on its ending lifeline; value of time should be greater than or equal to the value of the starting time, in terms of lifeline position
Message Is Associated With Action*		Input/Output information inherent in model logic	
	Message Name*		Label of the Message
	Action Name*		Name/identifier of an action
Message Is Associated with Information Exchange*		Input/Output information associated with OV-3 Information Exchange	
	Message Name*		Label of the message
	Information Exchange		Identifier of an information exchange associated with

Data Elements	Attributes	ARIS Mapping	Example Values/Explanation
	Identifier*		the message
Message Is Associated With Event*		Input/Output information inherent in model logic	
	Action Name*		Name/identifier of an action
	Event Name*		Name of an event

Table 4-9: Data Element Definitions for Logical Data Model, OV-7

Data Elements	Attributes	ARIS Mapping	Example Values/Explanation
Graphical Box Types			
Entity Type*		Entity Type	
	Name*	Attrib: Name	Name of the type of person, place, thing, or message of interest
	Description*	Attrib: Description/ Definition	Textual description of the entity type
Graphical Arrow Types			
Relationship Type*		Relationship Type	
	Name*	Attrib: Name	Name/identifier of the relationship type
	Description*	Attrib: Description/ Definition	Textual description of the relationship represented
	Source Entity Type Name*	Inherited by relationship "defines" between Entity Type and Relationship Type	Name of the entity type at the source of the relationship
	Target Entity Type Name*	Inherited by relationship "defines" between Entity Type and Relationship Type	Name of the entity type at the target of the relationship
	Cardinality Designation*	Attribute: Complexity of relationship "defines" between Entity Type and	Examples: one to one, one to many, etc.

Data Elements	Attributes	ARIS Mapping	Example Values/Explanation
		Relationship Type	
Category Relationship Type		Object: Generalization Type	
	Name	Attrib: Name	Name of the subtyping relationship
	Description	Attrib: Description/ Definition	Textual description of the subtype relationship represented
	Source Discriminated Entity Type Name	Inherited by relationship “is supertype of” between Entity Type and Generalization Type	Name of the supertype that is the source of the relationship
	Discriminant Attribute Type Name		Name of the attribute type that provides the discriminant for the entity type (should be an attribute associated with the entity)
	Number of Discriminant Values		Number of different subtypes (if known)
Non-graphical Types			
Attribute Type		Object: D attribute	
	Name	Attrib: Name	Name of attribute type
	Definition	Attrib: Description/ Definition	Definition of attribute
	Reference	Attrib: Source	Reference to accepted definition of attribute, if one exists) (e.g., DDDS reference)
Data Model Rule			
	Name		Name/identifier of rule
	Type		Examples: Null rule; child delete rule, child update rule
	Text		Text of rule
Data Domain		ERM Domain available	
	Name		Name of data domain
	Description		Textual description of data domain
	Range Constraint		Value range allowable for attributes in data domain
	Size Constraint		Maximum number of characters in display representation
Referenced Types			

Data Elements	Attributes	ARIS Mapping	Example Values/Explanation
Rule			See OV-6a Definition Table
Relationships			
Entity Type Is Described By Attribute Type		Relationship: "Is describing for" between Attribute and Entity Type	
	Entity Type Name	Inherited by relationship	Name of entity type
	Attribute Type Name	Inherited by relationship	Name of an associated attribute type
	Role of attribute	Attrib: Type	For example: Key, Foreign Key, Non-Key
Relationship Type Is Described By Attribute Type		Relationship: "Is describing for" between Attribute and Relationship Type	
	Relationship Type Name	Inherited by relationship	Name of a relationship type
	Attribute Type Name	Inherited by relationship	Name of an associated attribute type
Data Domain Constrains Values of Attribute Type			
	Data Domain Name		Name of data domain
	Attribute Type Name		Name of attribute type whose values are selected from the data domain
Relationship Type Has Rule			
	Relationship Type Name		Name of a relationship type
	Rule Name		Name/identifier of a rule associated with that relationship type
Entity Type Has Rule			
	Entity Type Name		Name of an entity type
	Rule Name		Name/identifier of a rule associated with that entity type
Attribute Type has Rule			
	Attribute Type Name		Name of an attribute type
	Rule Name		Name/identifier of a rule associated with an attribute
Data Model Rule Is Associated with A			

Data Elements	Attributes	ARIS Mapping	Example Values/Explanation
Structural Assertion Type Rule			
	Data Model Rule Name		Name of a data model rule
	Rule Name		Name/identifier of a structural assertion type rule associated with a data model rule
Category Relationship Type Has Destination Entity Type		Relationship “Is supertype/subtype of” between Entity Type and Generalization Type	
	Category Relationship Type Name	Attrib: Name	Name of subtyping relationship
	Destination Entity Type Name	Inherited by relationship “is supertype of” between Entity Type and Generalization Type	Name of entity type that is a subtype
	Discriminant Value		Value of the discriminant attribute that is associated with the entity subtype
Entity Is Related To Information Exchange*		See SV-6	
	Entity Type Name*		Name of an entity type
	Information Exchange Identifier*		Identifier of an Information Exchange
	Description of Relationship*		Text description of the relationship between the entity type and the information exchange (e.g., same, is subset of, contains)
Entity Is Related To Operational Activity I/O Information*		See SV-4	
	Entity Type Name*		Name of an entity type
	Operational Activity I/O Information Name*		An of the information associated with an activity input or output flow
	Description of Relationship*		Text description of the relationship between the entity and activity I/O (e.g.,

Data Elements	Attributes	ARIS Mapping	Example Values/Explanation
			same, is subset of, contains)

Data Element Definitions for Systems Interface Description, SV-1

Data Elements	Attributes	ARIS Mapping	Example Values/Explanation
Graphical Box Types			
Systems Node*		Object: Application System	(Appears in internodal and intranodal versions)
	Name*	Attrib: Name	Name of a systems node, usually denoting a facility where an operational node is located. If the corresponding operational node is designated as a physical one, then the systems node name and the physical operational node name may refer to the same entity (definition).
	Description*	Attrib: Description/ Definition	Text summary description of systems node's corresponding role or mission and associated resources (e.g., people, platforms, facilities, systems) that perform these roles or missions
	Location*	Relationship:	The location of the

Data Elements	Attributes	ARIS Mapping	Example Values/Explanation
		“is located at” between Application System and Location	systems node (i.e., the location where the systems are resident)
System*		Object: Application System	(Appears in all versions)
	Name*	Attrib: Name	Name/identifier of system or subsystem
	Description*	Attrib: Description/ Definition	Text summary of function or set of functions performed and constituent parts contained, e.g., shared DB
System Component		Application System	Software or hardware item (component) of a system (or subsystem). (Appears in intrasystem version and appears optionally on other versions)
	Name	Attrib: Name	Name/identifier of system component, including model/version number
	Type	Attrib: Short Description	Type of component, for example: hardware component; platform component

Data Elements	Attributes	ARIS Mapping	Example Values/Explanation
			(i.e., combined hardware and system software); system software; or application (i.e., mission unique) software
	Standard Used		Name/identifier of the standard used in the system component
	Description	Attrib: Description/ Definition	Text description of function(s) supported by system component
	Vendor/Source	Source	Source of system component
Graphical Arrow Types			
Interface*		Object: Cluster	(Required in internodal and intranodal versions)
	Name*	Attrib: Name	Name/identifier of interface
	Description*	Attrib: Description/ Definition	Text description of interface; may include a discussion of communications systems or communications systems elements involved as well as indications as to whether interface is two-way or one-way only
	Endpoint 1:	Inherited by	Name of entity

Data Elements	Attributes	ARIS Mapping	Example Values/Explanation
	System Name*	relationship “has output of” between Application System and Cluster	(systems node, system) that is at one end of the interface. In case of one-way connections, this endpoint is the source endpoint
	Endpoint 2: System Name*	Inherited by relationship “is input for” between Application System and Cluster	Name of entity (systems node, system) that is at the other end of the interface. In case of one-way connections, this endpoint is the target endpoint
	Key Interface (KI) Designation		If applicable
	Interface Control Document (ICD) Name	Can be realized with attribute: Link	If applicable: Name/reference to the Interface Control Document (ICD) that specified the KI (when developed)
	Key Interface Rationale		Text describing why this is a Key Interface if applicable
Non-graphical Types			
System Function (if used)		Object: Function	
	Name	Attrib: Name	Name/identifier of system function
	Description	Attrib:	Text summary

Data Elements	Attributes	ARIS Mapping	Example Values/Explanation
		Description/ Definition	description of system function
Referenced Types			
Operational Capability*		Object: Function	See OV-5 Definition Table
Operational Activity*		Object: Function	See OV-5 Definition Table
Needline*		Object: Cluster	See OV-2 Definition Table
Standard*			See TV-1 Definition Table
Relationships			
Systems Node Contains (Sub)-System(s)*		Relationship: “Subsumes” connection between Application Systems	
	Systems Node Name*	Attrib: Name	Name/identifier of a systems node
	System Name*	Attrib: Name	Name/identifier of a system resident at that node
System Contains (Sub)-System(s)		Relationship: “Subsumes” connection between Application Systems	
	Systems Name	Attrib: Name	Name/identifier of a system
	(Sub)-System(s) Name	Attrib: Name	Name/identifier of a subsystem contained within system
System Contains		Relationship:	

Data Elements	Attributes	ARIS Mapping	Example Values/Explanation
System Component		“Subsumes” connection between Application Systems	
	System Name	Attrib: Name	Name/identifier of a system
	System Component Name	Attrib: Name	Name/identifier of a component of the system
System Performs System Function*		Relationship: “Supports” connection between Application System and Function	
	System Name*	Attrib: Name	Name/identifier of a system
	System Function Name*	Attrib: Name	Name/identifier of a system function performed by system
Systems Node Supports Operational Node*		Relationship: “Supports” connection between Application System and Function	
	Systems Node Name*	Attrib: Name	Name/identifier of systems node
	Operational Node Name*	Attrib: Name	Name/identifier of operational node supported by automation resident at the systems node

Data Elements	Attributes	ARIS Mapping	Example Values/Explanation
System/Subsystem Supports Operational Node		Relationship: “Supports” connection between Application System and Function	
	System Name	Attrib: Name	Name/identifier of system
	Operational Node Name	Attrib: Name	Name/identifier of an operational node with activities supported by the system
System/Subsystem Supports Operational Capability*		Relationship: “Supports” connection between Application System and Function	
	System Name*	Attrib: Name	Name/identifier of system
	Operational Capability Name*	Attrib: Name	Name/identifier of an operational capability supported by the system
	Support Status Code		One of: red, yellow, or green. Red means capability planned but not developed; yellow means partial capability provided OR full capability provided but system has not

Data Elements	Attributes	ARIS Mapping	Example Values/Explanation
			been fielded; and green means full capability provided and system has been fielded
System/Subsystem Supports Operational Activity*		Relationship: "Supports" connection between Application System and Function	
	System Name*	Attrib: Name	Name/identifier of a system
	Operational Activity Name*	Attrib: Name	Name of an activity supported by the system
System Component Supports Operational Capability		Relationship: "Supports" connection between Application System and Function	
	System Component Name	Attrib: Name	Name/identifier of a system component
	Operational Capability Name	Attrib: Name	Name of an operational capability supported by the system component
System Component Supports Operational Activity		Relationship: "Supports" connection between Application System and	

Data Elements	Attributes	ARIS Mapping	Example Values/Explanation
		Function	
	System Component Name	Attrib: Name	Name/identifier of a system component
	Operational Activity Name	Attrib: Name	Name of an activity supported by the system component
Interface Implements Automated Portion of Needline*		See OV-3	
	Interface Name*		Name/identifier of interface
	Needline Name*		Name/identifier of needline partially or fully implemented by the Interface
System/Subsystem Uses Standard*			
	System Name*		Name/identifier of system
	Standard Name*		Name/ID number of a standard used by the system
System Component Uses Standard*			
	System Component Name*		Name/identifier of system component
	Standard Name*		Name/ID number of a standard used by the system component

Data Elements	Attributes	ARIS Mapping	Example Values/Explanation
Graphical Box Types			
Systems Node*		Object: Application System	See SV-1 Definition Table
System*		Object: Application System	See SV-1 Definition Table
Communications System*		Object: Network Node	
	Name*	Attrib: Name	Name/identifier of communications system whose primary function is to control the transfer and movement of data or information. Examples include network switches and routers and communications satellites
	Description*	Attrib: Description/ Definition	Text summary description of communications functions of the communications system
Graphical Arrow Types			
Communications Link*		Object: Network Connection	A communications link connects systems nodes or systems (including communications systems).

Data Elements	Attributes	ARIS Mapping	Example Values/Explanation
	Name*	Attrib: Name	Name/identifier of a communications link that describes a single physical connection to communicate from one systems node/system to another
	Communication Standard, Protocols Supported*	Attrib: Description/ Definition	For example, TCP/IP; Link-11
	Capacity*	Attrib: Transmission Speed	Throughput; channel capacity, bandwidth
	Infrastructure Technology*	Attrib: Transmission Medium	Infrastructure technology supporting this communications link; the level of detail contained here will depend on the purposes of the architecture (e.g., fiber, twisted pair, wireless, ATM, internet, intranet, Ethernet, VPN, radio plus frequency, encryption (if any))
	Endpoint 1: Systems Node/System Name*	Inherited by relationship "subsumes" between Network Node and Network Connection	Name of graphic box that is at one end of the communications link on the diagram; in case of one-way

Data Elements	Attributes	ARIS Mapping	Example Values/Explanation
			connections, this endpoint is the source endpoint. The endpoint of a communications link may also be listed as “External” if the endpoint is outside the scope of the architecture or diagram.
	Endpoint 2: Systems Node/System Name*	Inherited by relationship “subsumes” between Network Node and Network Connection	Name of the graphic box that is at the other end of the communications link on the diagram; in case of one-way connections, this endpoint is the destination endpoint. The endpoint of a communications link may also be listed as “External” if the endpoint is outside the scope of the architecture or diagram.
Non-graphical Types			
Pathway		Object: Network connection	A series of communications link that support an interface (from SV-1)
	Name	Attrib: Name	Name/identifier of

Data Elements	Attributes	ARIS Mapping	Example Values/Explanation
			a multiple communications link pathway that describes a single way (i.e., with no options) to communicate from one systems node/system to another
	Description	Attrib: Description/ Definition	Textual description of pathway, including whether the pathway is one-way only or two-way
	Endpoint 1 Systems Node/System Name	Inherited by relationship “subsumes” between Network Node and Network Connection	Name of systems node or system at one end of pathway; if pathway is one-way, this endpoint should be the source endpoint. May be listed as “External”
	Endpoint 2 Systems Node/System Name	Inherited by relationship “subsumes” between Network Node and Network Connection	Name of systems node or system at the other end of pathway; if pathway is one-way, this endpoint should be the destination endpoint. May be listed as “External”
	Number of communications Links	Attrib: Number of Channels	Number of communications links or steps in

Data Elements	Attributes	ARIS Mapping	Example Values/Explanation
			the pathway
LAN		Object: Network	
	Name	Attrib: Name	Name/identifier of local area network
	Description	Attrib: Description/ Definition	Textual description of LAN, including purpose, size, and capability
Network		Object: Network	A collection of communications links (and systems nodes and systems where applicable)
	Name	Attrib: Name	Name/identifier for a Wide Area Network or Metropolitan Area Network
	Description	Attrib: Description/ Definition	Textual description of network purpose, size, and capability
	Security Classification	Attrib: Processing code	Classification of data that the network is allowed to carry
Relationships			
LAN Contains communications Link		Relationship: "Belongs to" relationship between Network and Network Connection	
	LAN Name	Inherited by	Name/identifier of

Data Elements	Attributes	ARIS Mapping	Example Values/Explanation
		relationship	a LAN
	communications Link Name	Inherited by relationship	Name/identifier of a communications link that makes up part of the LAN
Systems Node Contains LAN		Indirect relationship between Application System, Organizational Unit and Network	
	Systems Node Name	Inherited by relationship	Name/identifier of a systems node
	LAN Name	Inherited by relationship	Name/identifier of a LAN contained within the systems node
Pathway Contains Communications Link(s)		Relationship “consists (vert.) of section” connection between Network Connection and Network Connection	
	Pathway Name	Inherited by relationship	Name/identifier of pathway
	Communications Link	Inherited by relationship	Name/identifier of the Communications link within the pathway
	Communications Link Position In	Attrib: Description/	Position of the Communications

Data Elements	Attributes	ARIS Mapping	Example Values/Explanation
	Pathway	Definition	link in the pathway, given in terms of number of Communications link from endpoint 1
Pathway Details Interface			
	Pathway Name		Name/identifier of pathway
	Interface Name		Name/identifier of an interface from SV-1 that is an abstraction of the pathway
Network Contains LAN		LAN is a network (can use “subsumes” connection for hierarchy)	
	Network Name	Inherited by relationship	Name/identifier of a network
	LAN Name	Inherited by relationship	Name/identifier of a LAN that is part of the network
Network Contains Communications Link		Relationship: “Belongs to” between Network Connection and Network	
	Network Name	Inherited by relationship	Name/identifier of a network
	Communications Link Name	Inherited by relationship	Name/identifier of a Communications link that is part of

Data Elements	Attributes	ARIS Mapping	Example Values/Explanation
			the network
Network Contains Communications System		Relationship: “Belongs to” between Network and Network Node	
	Network Name	Inherited by relationship	Name/identifier of a network
	Communications system Name	Inherited by relationship	Name/identifier of a communications system that is part of the network
System Is Attached to Network		Indirect relationship between Application System, Organizational Unit and Network	
	System Name		Name/identifier of a system
	Network Name		Name/identifier of a network to which the system is attached
Systems Node Is Attached to Network		Indirect relationship between Application System, Organizational Unit and Network	
	Systems Node Name		Name/identifier of a systems node
	Network Name		Name/identifier of

Data Elements	Attributes	ARIS Mapping	Example Values/Explanation
			a network that is attached to the node (i.e., a network to which all systems at the systems node are connected via a common service
Communications System Uses Standard*			
	Communications System Name*		Name/identifier of communications system
	Standard Name*		Name/ID number of a standard used by the communications system
Communications Link Uses Standard*			
	Communications Link Name*		Name/identifier of communications link
	Standard Name*		Name/ID number of a standard used by the communications link
Network Uses Standard*			
	Network Name*		Name/identifier of network
	Standard Name*		Name/ID number of a standard used by the network

Data Element Definitions for Systems-Systems Matrix, SV-3

Data Elements	Attributes	ARIS Mapping	Example Values/Explanation
Non-graphical Types			
Code Legend Type*		N/A	
	Name*		Name of the code legend type. Example: status, purpose, classification level, means
	Description*		Textual description of the code legend type used to categorize interface characteristics
Code Legend*		N/A	
	Code*		A unique code representing certain interface characteristic
	Type Name*		Name of a code legend type this code is categorized under
	Description*		Textual description of the interface characteristic represented by the code
System*		See SV-1 Definition Table	
Interface Entry*		Object: D Attribute	
	Name*	Attrib: Name	Name/identifier of interface entry. It should be the same as the name of the interface for which this interface entry is detailing the

Data Elements	Attributes	ARIS Mapping	Example Values/Explanation
			characteristics of the interface.
	Description*	Attrib: Description/ Definition	Textual summary description of the interface entry
Referenced Types			
Interface*		Object: Cluster	See SV-1 Definition Table
Relationship			
Code Legend Represents Characteristics Of Interface Entry			
	Code Legend Code		Code of a code legend, which implies the code legend type. (Note that the characteristics of an interface entry can not be represented by more than one code legend with the same code legend type.)
	Interface Entry Name		Name/identifier of an interface entry for which the characteristics of the interface entry are represented by the code
Interface Entry Details Interface		Relationship "Consists of" connection between Cluster and D Attribute	
	Interface Entry Name	Inherited by relationship	Name/identifier of interface entry
	Interface	Inherited by	Name/identifier of a

Data Elements	Attributes	ARIS Mapping	Example Values/Explanation
	Name	relationship	interface for which the interface entry is detailing its characteristics

Data Element Definitions for Systems Functionality Description, SV-4

Data Elements	Attributes	ARIS Mapping	Example Values/Explanation
Graphical Box Types			
System Function*		Object: Function	See SV-1 Definition Table
External Data Source/Sink*		Object: Cluster	See SV-1 Definition Table (system node object)
	Name*	Attrib: Name	Name/identifier for a data source or sink (e.g., node, system, organization, or user) outside the scope of the architecture
	Description*	Attrib: Description/ Definition	Textual description of the external data source or sink
Data Repository*/ Shared Database		Object: Application System	
	Name*	Attrib: Name	Name/identifier of data store
	Description*	Attrib: Description/ Definition	Textual summary description of data store
Graphical Arrow Types			

Data Elements	Attributes	ARIS Mapping	Example Values/Explanation
Data Flow*		Entity Type	Same as Data Exchange in SV-6
	Name*	Attrib: Name	Name/identifier of data flow (may be the same as a Systems Data Exchange name)
	Description*	Attrib: Description/ Definition	Textual description of the data flow
	“From” System Function/External Data Source/Data Repository*	Inherited by relationship “has output of” from Function to Entity Type	Name of box entity from which the arrow originates
	“To” System Function/External Data Sink/Data Repository*	Inherited by relationship “is input for” from Entity Type to Function	Name of box entity at which the arrow terminates
Function Decomposition Connector		Relationship: “Process-oriented superior” connection between Functions.	
	Super Function	Inherited by relationship	Name/Identifier of function that is being decomposed
	Sub-Function	Inherited by relationship	Name/Identifier of system sub-function into which the super-function decomposes

Data Elements	Attributes	ARIS Mapping	Example Values/Explanation
Referenced Types			
Systems Node		Object: Application System	See SV-1 Definition Table
System (or subsystem)*		Object: Application System	See SV-1 Definition Table
System Component		Object: Application System	See SV-1 Definition Table
Interface*		Object: Cluster	See SV-1 Definition Table
Relationships			
Data Flow Is Associated With Systems Data Exchange		Relationship: “Has relationship to” connection between Entity Types	See SV-6 Definition Table
	Data Flow Name	Inherited by relationship	Name of a data flow
	Systems Data Exchange Name	Inherited by relationship	Name/identifier of Systems Data Exchange associated with the data flow
Data Flow Is Involved in Interface*		Indirect relationship: “Consists of” connection between Cluster (Interface) and Entity Type (System Data	See SV-6 Definition Table

Data Elements	Attributes	ARIS Mapping	Example Values/Explanation
		Exchange) and “Has relationship to” connection between Entity Types (System Data Exchange and Data Flow)	
	Data Flow Name	Inherited by relationship	Name of a data flow
	Interface Name	Inherited by relationship	Name/identifier of an interface (from SV-1)
Data Repository Is Sink For Data Flow		Data Repository is represented by the Application System that manages the database. Application System (the sink) is shown as “supporting” the corresponding system Function that has the Data Flow as an input.	
	Data Repository Name	Inherited by relationship	Name/identifier of a data store
	Data Flow Name	Inherited by relationship	Name/identifier of a data flow that is

Data Elements	Attributes	ARIS Mapping	Example Values/Explanation
			input to the data store
Data Repository Is Source For Data Flow		Data Repository is represented by Application System that manages the database. Application System (the sink) is shown as “supporting” the corresponding system function that has the data flow as an output.	
	Data Repository Name	Inherited by relationship	Name/identifier of a data store
	Systems Data Exchange Name	Inherited by relationship	Name/identifier of a data flow that is output from the data store
External Sink Is Sink For Data Flow		Data Repository is represented by Application System that manages the database. Application System (the sink) is shown as supporting the	

Data Elements	Attributes	ARIS Mapping	Example Values/Explanation
		corresponding system Function that has the Data Flow as an input	
	External Sink Name	Inherited by relationship	Name/identifier of an external sink
	Data Flow Name	Inherited by relationship	Name/identifier of a data flow that is input to the data store
External Source Is Source For Data Flow		Data Repository is represented by Application System that manages the database. Application System (the sink) is shown as supporting the corresponding system Function that has the Data Flow as an output.	
	External Source Name	Inherited by relationship	Name/identifier of an external source
	Systems Data Exchange Name	Inherited by relationship	Name/identifier of a data flow that is output from the data store
System Function		Relationship: "Has output"	

Data Elements	Attributes	ARIS Mapping	Example Values/Explanation
Produces Data Flow		of” between Function and Entity Type	
	System Function Name	Inherited by relationship	Name/identifier of system function
	Systems Data Exchange Name	Inherited by relationship	Name/identifier of data flow that is output from the system function
System Function Processes (consumes) Data Flow		Relationship: “Is input for” between Entity Type and Function	
	System Function Name	Inherited by relationship	Name/identifier of system function
	Systems Data Exchange Name	Inherited by relationship	Name/identifier of data flow that is input to the system function
System Function Is Allocated To Systems Node		Relationship: “Supports” between Application System and Function	
	System Function Name	Inherited by relationship	Name/identifier of system function
	Systems Node Name	Inherited by relationship	Name/identifier of systems node to which the Function has been allocated
System Function Is Allocated To System*		Relationship: “Supports” between Application System and	

Data Elements	Attributes	ARIS Mapping	Example Values/Explanation
		Function	
	System Function Name	Inherited by relationship	Name/identifier of system function
	System Name	Inherited by relationship	Name/identifier of system to which the function has been allocated
System Function Is Allocated To Systems Component		Relationship: “Supports” relationship between Application System and Function	
	System Function Name	Inherited by relationship	Name/identifier of system function
	Systems Component Name	Inherited by relationship	Name/identifier of systems component to which the function has been allocated
System Contains Data Repository		Attrib: database management system	
	System Name		Name/identifier of system
	Data Repository Name		Name/identifier of data store name contained in the system
System Function Uses Standard*			
	System Function Name*		Name/identifier of system function
	Standard*		Name of JTA

Data Elements	Attributes	ARIS Mapping	Example Values/Explanation
			standard

Data Element Definitions for Operational Activity to System Function Matrix, SV-5

Data Elements	Attributes	ARIS Mapping	Example Values/Explanation
Referenced Types			
Operational Capability*			See OV-5 Definition Table
System*			See SV-1 Definition Table
Operational Activity*			See OV-5 Definition Table
System Function*			See SV-4 Definition Table
Relationships			
System Supports Operational Capability		Relationship “Supports” between Application System and Function	
	System Name	Inherited by relationship	Name/identifier of system
	Operational Capability Name	Inherited by relationship	Name/identifier of an operational capability supported by the system
	Support Status Code	Attrib: Remark/Example of Relationship	One of: red, yellow, or green. Red means capability planned but not developed; yellow means partial capability provided or full capability provided but system has not been

Data Elements	Attributes	ARIS Mapping	Example Values/Explanation
			fielded; and green means full capability provided and system has been fielded
System Implements Operational Capability*		Relationship “Supports” between Application System and Function	
	System Name*	Inherited by relationship	Name/identifier of system
	Operational Capability Name*	Inherited by relationship	Name/identifier of operational capability implemented by the system
Operational Activity Is Supported By System Function*		This relationship is inherent in the model because we have defined system function as an operational activity supported by a system.	
	Operational Activity Name*		Name/identifier of operational activity
	System Function Name*		Name/identifier of system function that supports the operational activity
System Function Implements Operational Activity*		This relationship is inherent in the model because we have defined system function	

Data Elements	Attributes	ARIS Mapping	Example Values/Explanation
		as an operational activity supported by a system.	
	System Function Name*		Name/identifier of system function
	Operational Activity Name*		Name/identifier of operational activity (at least partially) implemented by the system function

Data Element Definitions for Systems Data Exchange Matrix, SV-6

Data Elements	Attributes	ARIS Mapping	Example Values/Explanation
Non-graphical Types			
Systems Data Exchange*		Object: Entity Type	
	Identifier*	Attrib: Identifier	Identifier of Systems Data Exchange – usually based on the relevant needline, system interface, and/or information exchange
	Name*	Attrib: Name	Name of Systems Data Exchange – usually based on the relevant

Data Elements	Attributes	ARIS Mapping	Example Values/Explanation
			needline, system interface, and/or information exchange
	Content Description*	Attrib: Description/Definition	Definition of data exchange
	Format Type*	Attrib: Short Description	Application level format (e.g., XML/DTD, EDI, ASCII Text) with parameters and options used, or other relevant protocol
	Media Type*	Attrib: Application	Type of media
	Data Standard*		Application level format (e.g., XML/DTD, EDI, ASCII Text) with parameters and options used, or other relevant protocol
	Units of Measurement*		Units used for data
	Accuracy		Accuracy of information
	Transaction Type*	Attrib: Processing Code	Descriptive field that identifies the type of the information exchange. Valid values are: publish, search, subscribe, collaborate
	Triggering Event*	Relationship: "is state of" Relation	Brief textual description of the

Data Elements	Attributes	ARIS Mapping	Example Values/Explanation
		between Event and Cluster in OV-6	event(s) that triggers the System exchange.
	Interoperability Level Achievable		Level of Information Systems Interoperability (LISI)
	Sending System Name*		Name/identifier of system that generates the data exchange
	Sending System Function*		(if used) The identity of the System Function generating the exchange.
	Receiving System Name*		Name/identifier of the system that receives the data exchange
	Receiving System Function*		The identity of the System Function receiving the exchange.
	Periodicity*		Frequency of data exchange transmission – may be expressed in terms of worst case or average frequency.
	Timeliness*		How much delay this data exchange can tolerate and still be relevant to the receiving system. This may

Data Elements	Attributes	ARIS Mapping	Example Values/Explanation
			be an allocation of the lifeline factor from the related information exchange
	Throughput*		Bits or bytes per time period – may be expressed in terms of maximum or average throughput required
	Size*		Size of data
	IA – Access Control Type Code		The code that represents the class of mechanisms used to ensure only those authorized can access information for a specific information-exchange-requirement-assurance
	IA – Availability Effort Code		The code that represents the relative level of effort required to be expended to ensure that the information can be accessed for use in a specific information-exchange-requirement-assurance

Data Elements	Attributes	ARIS Mapping	Example Values/Explanation
	IA – Confidentiality Type Code		The code that represents the kind of protection required for information to prevent unintended disclosure in a specific information-exchange-requirement-assurance
	IA – Dissemination Control Type Code		The code that represents the kind of restrictions on receivers of information for a specific information-exchange-requirement-assurance based on sensitivity of information
	IA – Information Criticality Code		The code that represents the seriousness of the benefit that the information exchange element provides to the objective of the action being taken for a specific information-exchange-requirement-assurance

Data Elements	Attributes	ARIS Mapping	Example Values/Explanation
	IA – Integrity Type Code		The code that represents the kind of requirements for checks that the content of the data element of a specific information-exchange-requirement-assurance has not been altered
	IA – Non-Repudiation Consumer Code		The code that represents the requirements for unassailable knowledge that the information sent was consumed by the intended recipient for a specific information-exchange-requirement-assurance
	IA – Non-Repudiation Producer Code		The code that represents the requirements for unassailable knowledge that the information received was produced by the stated source for a specific information-exchange-requirement-

Data Elements	Attributes	ARIS Mapping	Example Values/Explanation
			assurance
	IA - Protection Duration Code		The code that represents how long the information must be safeguarded for a specific information-exchange-requirement-assurance.
	IA - Protection Elapsed-Time Quantity		The elapsed-time quantity that represents the length of time during which information assurance protections of the information is required for a specific information-exchange-requirement-assurance.
	IA - Protection Suspense Calendar Date		The calendar date on which the designated level of safeguarding discontinues for a specific information-exchange-requirement-assurance.
	Classification*		Classification Code for the Information

Data Elements	Attributes	ARIS Mapping	Example Values/Explanation
			Element
	Classification Caveat*		A set of restrictions on information of a specific classification. Supplements a security classification with information on access, dissemination, and other types of restrictions
	Releasability Code		The code that represents the kind of controls required for further dissemination of information for a specific information-exchange-requirement-assurance
Referenced Types			
Needline*			See OV-2 Definition Table
System Interface*			See SV-1 Definition Table
Information Exchange*			See OV-3 Definition Table
Relationships			
Data Exchange Implements Automated		Relationship “Consists of” between Cluster and Entity Type	(For Systems Data Exchanges that are directly related to operational

Data Elements	Attributes	ARIS Mapping	Example Values/Explanation
Portion of Information Exchange*			information exchanges)
	Data Exchange Identifier*	Inherited by relationship	Identifier for a Systems Data Exchange
	Information Exchange Identifier*	Inherited by relationship	Identifier for an automated operational information exchange (from OV-3)
Data Exchange Is Carried by Interface*		Relationship “Consists of” between Cluster and Entity Type	
	Data Exchange Identifier*	Inherited by relationship	Identifier for a Systems Data Exchange
	Interface Name*	Inherited by relationship	Name/identifier for an interface (from SV-1)
Data Exchange Uses Standard*			
	Data Exchange Identifier*		Identifier for a Systems Data Exchange
	Standard Name*		Name/ID number of a standard used by the system data exchange

Data Element Definitions for Systems Performance Matrix, SV-7

Data Elements	Attributes	ARIS Mapping	Example Values/Explanation
Non-graphical Types			
Performance Parameter Set*		Same as Performance Parameter Type.	
	Name*		Name/identifier of parameter set
	Number of performance parameters in set*		Number of different performance parameters for which measures were/will be taken
Performance Parameter Type*		Object: Key Performance Indicator	
	Name*	Attrib: Name	Name/identifier of performance parameter type (e.g., mean time between failures, maintainability, availability, system initialization time, data transfer rate, program restart time for platforms; and data throughput/capacity; response time, effectiveness, mean time between software failures for application software)
	Description*	Attrib: Description/Definition	Textual description of the performance parameters and what their

Data Elements	Attributes	ARIS Mapping	Example Values/Explanation
			measurements mean
Measurement Time*			
	ID*	Attrib: Processing Code	Positive integer less than or equal to the Number of Measurements shown in the product. This attribute identifies the position of the column of measurements (i.e., first column of measurements, second column, etc.)
	Timestamp*	Attrib: User attribute Point in time	Date when the measurements were/will be taken
Performance Measurement*			
	Performance Parameter Type Name*	Attrib: Name	Name/identifier of a performance parameter type
	Measurement Time ID*	Attrib: Processing Code	Positive integer identifying the position of the column of (past) measurements.
	Expected Value*	Attrib: Actual Value	Value of performance parameter type that was measured at the indicated time
Performance Goal*		Object: Objective	

Data Elements	Attributes	ARIS Mapping	Example Values/Explanation
	Performance Parameter Type Name*	Inherited by relationship	Name/identifier of a performance parameter type
	Measurement Time ID*	Attrib: Processing Code	Positive integer identifying the position of the column of (future) measurements
	Measurement Value*	Attrib: Plan Value	Value of performance parameter type that is expected at the indicated time
Referenced Types			
System*			See SV-1 Definition Table
System Component*			See SV-1 Definition Table
System Interface*			See SV-1 Definition Table
System Function*			See SV-4 Definition Table
Timed Technology Forecast			See SV-9 Definition Table
Milestone			See SV-8 Definition Table
Relationships			
System Contains System Component*			See SV-1 Definition Table
System Has Performance Parameter		Relationship: “Has output of” between application system	

Data Elements	Attributes	ARIS Mapping	Example Values/Explanation
Set*		and KPI	
	System Name*	Inherited by relationship	Name/Identifier of a system
	Performance Parameter Set Name*	Inherited by relationship	Name of the performance parameter set.
System Component Has Performance Parameter Set*		Relationship: “Has output of” between application system and KPI	
	System Component Name*	Inherited by relationship	Name/identifier for system component
	Performance Parameter Set Name*	Inherited by relationship	Name of the performance parameter set
System Interface Has Performance Parameter Set*		Relationship: “IS input for” between Cluster and KPI	
	System Interface Name*	Inherited by relationship	Name/identifier for system Interface
	Performance Parameter Set Name*	Inherited by relationship	Name of the performance parameter set
System Function Has Performance Parameter Set*		Relationship “supports” between Objective and Function and “is measured by” between Objective and KPI	
	System	Inherited by	Name/identifier for

Data Elements	Attributes	ARIS Mapping	Example Values/Explanation
	Function Name*	relationship	system function
	Performance Parameter Set Name*	Inherited by relationship	Name of the performance parameter set
Performance Parameter Set Includes Performance Measurement*		Attribute information of KPI	
	Performance Parameter Set Name*		Name/identifier of performance parameter set
	Performance Measurement Name*		Name/identifier of performance measurement to be included in performance parameter set
Performance Parameter Set Includes Performance Goal*		Relationship “is measured by” between Objective and KPI	
	Performance Parameter Set Name*	Inherited by relationship	Name/identifier of performance parameter set
	Performance Goal Name*	Inherited by relationship	Name/identifier of performance goal to be included in performance parameter set
Timed Technology Forecast Is Related To Performance Goal			

Data Elements	Attributes	ARIS Mapping	Example Values/Explanation
	Timed Technology Forecast Name*		Name/identifier of a forecast regarding a specific technology
	Performance Goal Name*		Name of the performance goal related to the timed technology forecast
Performance Goal Supports Evolution Milestone			
	Performance Goal Name*		Name of the performance goal supporting the milestone
	Milestone Name*		Name/identifier of a system evolution milestone

Data Element Definitions for Systems Evolution Description, SV-8

Data Elements	Attributes	ARIS Mapping	Example Values/Explanation
Non-graphical Types			
System*			See SV-1 Definition Table
System Component			See SV-1 Definition Table
Timeline*			
	Name*		Name of timeline
	Description*		Textual description of purpose of timeline, including goals of the migration or evolution

Data Elements	Attributes	ARIS Mapping	Example Values/Explanation
			process being documented
	Beginning Time*		Date of beginning of timeline
	Ending Time*		Date of ending of timeline
Milestone*			
	Name*		Name/identifier for milestone
	Date*		Date for achieving milestone in terms of month and year or number of months from baseline date
	Description*		Goals to be achieved at milestone
Graphical Arrow Types			
System Grouping Link*			
	Milestone Name*		Name/identifier of the milestone when this grouping should complete integration
	System Group Name*		Name/identifier for a set of systems, subsystems, or system components
	Number of Constituent systems, subsystems or System Components*		Number of systems, subsystems, or system components grouped together
Non-graphical Types			
System			

Data Elements	Attributes	ARIS Mapping	Example Values/Explanation
Group*			
	Name*		Name/identifier for a set of systems, subsystems, or system components
	Description*		Textual description of system group
Referenced Types			
Timed Technology Forecast*			See SV-9 Definition Table
Standard*			See TV-1 Definition Table
Timed Standards Forecast*			See TV-2 Definition Table
Relationships			
System Group Contains Constituent System, Subsystem, or System Component*			
	System Group Name*		Name/identifier for a set of systems, subsystems, or system components
	System, subsystem, or System Component Name*		Name of systems, subsystems, or system components contained in the system group
Milestone Is Completed With System Group*			
	Milestone		Name/identifier of

Data Elements	Attributes	ARIS Mapping	Example Values/Explanation
	Name*		milestone
	System Group Name*		Name/identifier for a set of upgraded systems, subsystems, or system components required to complete a milestone
	System Version*		Version name/number for system configuration at the completion of milestone
Timeline Has Beginning System Configuration *			
	Timeline Name*		Name/identifier of timeline
	System Name*		Name of a system (for system evolution timelines)
	System Version*		Version name/number for system configuration at the beginning of timeline (for system evolution)
Timeline Has Ending New System*			
	Timeline Name*		Name/identifier of timeline
	System Name*		Name of new system (or subsystems) available at end of timeline
Timeline Contains Milestone*			
	Timeline Name*		Name/identifier of timeline

Data Elements	Attributes	ARIS Mapping	Example Values/Explanation
	Milestone Name*		Name/identifier of milestone
	Relative Position of Milestone*		Position of milestone on timeline relative to beginning of timeline (e.g., first, fifteenth)
Milestone Requires Timed Technology Forecast			
	Milestone Name		Name/identifier of milestone
	Timed Technology Forecast Name		Name/identifier of a timed technology forecast for a specific technology required for supporting the system associated with the milestone. That is, milestone cannot be met if technology forecasted is not available
Milestone Includes Timed Standards Forecast			
	Milestone Name		Name/identifier of milestone
	Timed Standards Forecast Name		Name/identifier of a Timed Standards Forecast for a specific standard that will be used in the system version associated with the milestone.
Milestone Includes Upgrade to			

Data Elements	Attributes	ARIS Mapping	Example Values/Explanation
Existing Standard			
	Milestone Name		Name/identifier of milestone
	Standard Name		Name/identifier of a standard from TV-1 Technical Standards Profile that will be used in system, subsystem, or system components integrated for the milestone

Data Element Definitions for Systems Technology Forecast, SV-9

Data Elements	Attributes	ARIS Mapping	Example Values/Explanation
Non-graphical Types			
Technologies Forecast Profile*			
	Name*		Name/identifier of Technologies Forecast Profile
	Description*		Textual description of purpose of forecast
Timed Technology Forecast*			
	Name*		Name/identifier of a forecast regarding a specific technology
	Technology*		Textual description of a specific technology being forecasted
	Time frame*		Time frame for which

Data Elements	Attributes	ARIS Mapping	Example Values/Explanation
			forecast is valid; usually expressed in terms of a (future) date or months from baseline
	Discussion*		Textual notes regarding technology status, likely commercial market acceptance, and risk assessment of adopting the technology forecasted
	Confidence Factor*		Textual description of confidence level in forecast
Referenced Types			
Reference Model			See TV-1 Definition Table
Service Area			See TV-1 Definition Table
Service			See TV-1 Definition Table
System			See SV-1 Definition Table
System Component			See SV-1 Definition Table
Communications System			See SV-2 Definition Table
Communications Link			See SV-2 Definition Table
Network			See SV-2 Definition Table
Relationships			
Technologies			

Data Elements	Attributes	ARIS Mapping	Example Values/Explanation
Forecast Profile Covers Service Area/Service			
	Technologies Forecast Profile Name		Name/identifier of technologies forecast profile
	Service Area/Service Name		Name/identifier of a TRM service area or service covered by the technologies forecast profile
Service Area/Service Has Timed Technology Forecast			
	Service Area/Service Name		Name/identifier of a service area or service
	Timed Technology Forecast Name		Name/identifier of a specific, time sensitive forecast for technology relevant to the service area or service
Technologies Forecast Profile Is Based On Reference Model			
	Technologies Forecast Profile Name		Name/identifier of the Technologies Forecast Profile
	Reference Model Name		Name/identifier of the reference model used to organize the forecast
Timed			

Data Elements	Attributes	ARIS Mapping	Example Values/Explanation
Technology Forecast Impacts System			
	Timed Technology Forecast Name		Name/identifier of a timed technology forecast
	System		Name/identifier of a system that could be impacted by the technology forecast
Timed Technology Forecast Impacts System Component			
	Timed Technology Forecast Name		Name/identifier of a timed technology forecast
	System Component		Name/identifier of a system component that could be impacted by the technology forecast
Timed Technology Forecast Impacts communications Link			
	Timed Technology Forecast Name		Name/identifier of a timed technology forecast
	Communications Link Name		Name/identifier of a communications link that could be impacted by the technology forecast

Data Elements	Attributes	ARIS Mapping	Example Values/Explanation
Timed Technology Forecast Impacts Network			
	Timed Technology Forecast Name		Name/identifier of a timed technology forecast
	Network Name		Name/identifier of a network that could be impacted by the technology forecast
Timed Technology Forecast Impacts Communications System			
	Timed Technology Forecast Name		Name/identifier of a timed technology forecast
	Communications system Name		Name/identifier of a communications system (e.g., gateway, router, satellite) that could be impacted by the technology forecast

Data Element Definitions for Systems Rules Model, SV-10a

Data Elements	Attributes	ARIS Mapping	Example Values/Explanation
Non-graphical Types			
Rule*			See OV-6a Definition Table
Formal			See OV-6a Definition Table

Data Elements	Attributes	ARIS Mapping	Example Values/Explanation
Language*			
Referenced Types			
System			See SV-1 Definition Table
System Component			See SV-1 Definition Table
System Function			See SV-1 Definition Table
Relationships			
Rule Applies to System		Inherent in model logic	
	Rule Name		Name of an action assertion or derivation rule
	System Name		Name/identifier of a system or to which the rule applies
Rule Applies to System Component		Inherent in model logic	
	Rule Name		Name of an action assertion or derivation rule
	System Component Name		Name/identifier of a system component to which the rule applies
Rule Applies to System Function		Inherent in model logic	
	Rule Name		Name of an action assertion or derivation rule
	System Function Name		Name of a system function to which the rule applies

Data Element Definitions for Systems State Transition Description, SV-10b

Data Elements	Attributes	ARIS Mapping	Example Values/Explanation
Graphical Box Types			
State*			See OV-6b Definition Table
Graphical Arrow Types			
Transition*			See OV-6b Definition Table
Non-graphical Types			
State Chart*			See OV-6b Definition Table
Action*			See OV-6b Definition Table
Event*			See OV-6b Definition Table
Event Qualifier Attribute			See OV-6b Definition Table
Event Qualifier Action*			See OV-6b Definition Table
Event Qualifier Guard*			See OV-6b Definition Table
Event Qualifier Export Event			See OV-6b Definition Table
Referenced Types			
System Function			See SV-1 Definition Table
Rule			See OV-6a Definition Table
Relationships			
State Chart Has Terminal State*			See OV-6b Definition Table

Data Elements	Attributes	ARIS Mapping	Example Values/Explanation
Event Triggers Transition*			See OV-6b Definition Table
Transition Has Event Qualifier Attribute			See OV-6b Definition Table
Transition Has Event Qualifier Action*			See OV-6b Definition Table
Transition Has Event Qualifier Guard*			See OV-6b Definition Table
Transition Has Event Qualifier Export Event			See OV-6b Definition Table
State Has Associated Action*			See OV-6b Definition Table
Transition Has Associated Action*			See OV-6b Definition Table
Event Qualifier Guard Is Associated With Rule			See OV-6b Definition Table
Splitting Transition Has Ending State			See OV-6b Definition Table
Synchronizing Transition Has Starting State			See OV-6b Definition Table
Nesting State Has Contained State Chart			See OV-6b Definition Table

Data Elements	Attributes	ARIS Mapping	Example Values/Explanation
Concurrent Superstate Has Partition State Chart			See OV-6b Definition Table
Splitting Transition Has Corresponding Synchronizing Transition			See OV-6b Definition Table
Action Is Associated With System Function		Inherent in definition of system function.	
	Action Name		Name/identifier of an action
	System Function Name		Name/identifier of a system function
	Relationship Description		Text description of the relationship (e.g., action is same as system function, action is contained in system function, action contains system function)
Event Qualifier Action Is Related To System Function			See OV-6 Definition Table
	Event Qualifier Action Name		Name of an event qualifier action
	System Function Name		Name/identifier of a system function

Data Elements	Attributes	ARIS Mapping	Example Values/Explanation
	Relationship Description		Text description of the relationship (e.g., action is same as system function, action is contained in system function, action contains system function)

Data Element Definitions for Systems Event-Trace Description, SV-10c

Data Elements	Attributes	ARIS Mapping	Example Values/Explanation
Graphical Box Types			
Node Lifeline*			See OV-6c Definition Table
	Node Lifeline Name*		Name of the node lifeline
	Description*		Text description of any assumptions or scope constraints on the lifeline
System Function Lifeline*			See OV-6c Definition Table
	Function Lifeline Name*		Name of the lifeline
	Description*		Text description of any assumptions or scope constraints on the lifeline
Graphical Arrow Types			
Message*			See OV-6c Definition Table
Non-graphical Types			
Message			See OV-6c Definition Table

Data Elements	Attributes	ARIS Mapping	Example Values/Explanation
Time*			
Referenced Types			
Systems Node*			See SV-1 Definition Table
System*			See SV-1 Definition Table
Systems Component			See SV-1 Definition Table
System Function			See SV-4 Definition Table
Action*			See OV-6b Definition Table
Event*			See OV-6b Definition Table
Systems Data Exchange*			See SV-6 Definition Table
Relationships			
Lifeline Represents A Systems node, a system, a systems component, or a system function		Application System; Function	
	Lifeline Name		Name of the lifeline
	System Name		Name of the system represented by the lifeline
Message Starts At Time*			See OV-6c Definition Table
Message Ends At Time*			See OV-6c Definition Table
Message Is Associated			See OV-6c Definition Table

Data Elements	Attributes	ARIS Mapping	Example Values/Explanation
with Action*			
Message Is Associated With Event*			See OV-6c Definition Table
Message Is Associated With Systems Data Exchange*		Inherent in model logic	
	Message Name*		Label of the Message
	Systems Data Exchange Identifier*		Name/identifier of Systems Data Exchange

Data Element Definitions for Physical Schema, SV-11

Data Elements	Attributes	ARIS Mapping	Example Values/Explanation
Non-graphical Types			
Physical Schema Model*			
	Name*		Name/identifier of physical schema model
	Description*		Textual summary description of the mechanisms used to implement the logical data model; may include several different types of mechanisms and their associated models. For example, both messages and flat files may be used.

Data Elements	Attributes	ARIS Mapping	Example Values/Explanation
	Number of Component Models*		Number of other types of models that make up the physical schema model
Data/Message Model			
	Data/Message Standard Name		Name/identifier of data or messaging standard to be used (e.g., USMTF; TADIL A, B, J)
	Data/Message Format Name		Name/identifier of data or message format used within the message standard
	Data/Message Type Parameters/Options		Parameter and option values necessary to completely identify data or message format to be used
File Structure Model			
	File Name		Name/identifier of file used to hold data/information
	File Structure Type		Type of file structure used; this will vary by platform type (e.g., UNIX file; VSAM or FTAM for IBM/MVS platforms)
	Description		Textual or code description of record structure(s) within the file
Entity Relationship Diagram (ERD) Model			
	ERD Name		Name/identifier of specific entity-relationship model
	ERD Type		Name of specific form of notation used (e.g.,

Data Elements	Attributes	ARIS Mapping	Example Values/Explanation
			IDEF1X); may be tool dependent (e.g., DIAD, System Architect)
	Softcopy Reference		Location and file format for softcopy of the specific model (e.g., URL)
Data Definition Language (DDL) Model			
	DDL Name		Name/identifier of DDL schema or file
	DDL Language Type		Name of language in which the DDL is written (.e.g., SQL)
	Softcopy Reference		Location and file format for the softcopy of the DDL (e.g., URL)
Relationships			
Physical Schema Model Contains Data/Message Model*			
	Physical Schema Model Name*		Name/identifier of physical schema model
	Message Model/File Structure Model/ER D Model/DD L Model Name*		Name/identifier of one of the types of models that makes up the physical schema model

Data Elements	Attributes	ARIS Mapping	Example Values/Explanation
Logical Model Maps to Physical Schema Model			
	Logical Model Name		Name/Identifier of logical data model (see OV-7)
	Physical Schema Model Name		Name/Identifier of corresponding physical schema model
	Reference to Mapping Document		Location of hardcopy or softcopy of document containing the detailed mapping between the logical and physical schema models; there is no generic form for this mapping - it can be complex and varies based on the types of physical models used
Physical Schema Model Uses Standard			
	Physical Schema Model Name*		Name/identifier of physical schema model
	Standard name		JTA standard

Appendix B – ARIS Object Attribute Standards

Model Attributes

Attribute name	Opt/Req.	<i>Notes</i>
Model Type - Attributes		
Name	R	Name
Description/Definition	R	Purpose, Description
Type	A	Type

Function Attributes

Attribute Name	Opt./Req.	<i>DoDAF Attribute</i>
Function Attributes		
Name	R	Name
Identifier	O	Identifier
Description/Definition	R	Description
Hierarchy Number	O	Level Identifier
Short Description	O	Line of Business

DoDAF Discriminator	O	Explains DoDAF type information
System Attributes		
Title 1	O	Reference
Link 1	O	Reference
Parameter 1		
Transaction Code		
Attributes of external systems/SAP Attributes		
SAP Function Type	R	Level Identifier (SAP ERP only)
Cost		
Organizational Unit Attributes		
Total Cost		
Avg.		
Attribute Name	Opt./Req.	<i>DoDAF Attribute</i>
Organizational Unit Attributes		
Name	R	Name
Identifier	O	
Description/Definition	R	Description

Hierarchy Number	O	Level Identifier
Short Description	O	(Military) Service, e.g. Army, Navy
DoDAF Discriminator	R	Explains DoDAF type information

Organizational Unit Type Attributes

Attribute Name	Opt./Req.	<i>DoDAF Attribute</i>
Organizational Unit Type Attributes		
Name	R	Name
Identifier	O	Identifier
Description/Definition	R	Description
Hierarchy Number	O	Level Identifier
Short Description	O	(Military) Service, e.g. Army, Navy
DoDAF Discriminator	R	Explains DoDAF type information

Cluster Attributes

Attribute Name	Opt./Req.	<i>DoDAF Attribute</i>

Cluster Attributes		
Name	R	Name OPTIONAL Identifier
Identifier	R	Identifier
Description/Definition	R	Description
Processing Code	R	Transaction Type
DoDAF Discriminator	R	Explains DoDAF type information

No. of occurrences

Person Type Attributes

No. of occu

Attribute Name	Opt./ Req.	<i>DoDAF Attribute</i>
Person Type Attributes		
Name	R	Name
Description/Definition	R	Description

DoDAF Discriminator	R	Explains DoDAF type information

Connection Attributes

Attribute Name	Opt./ Req.	<i>DoDAF Attribute</i>
Connection Attributes		
Connection role	R	Name/Label
Description/Definition	R	Description
Type	A	Relationship Type
DoDAF Discriminator	R	Explains DoDAF type information
Complexity	O	Cardinality Designation (for “defines” relationship between entity Type and Relationship Type in ERM model)
Description/Definition	O	Communications Link Position In Pathway (for “consists (vert.) of section” relationship)
Remark/Example	O	Support Status Code (for “supports” relationship between Application System and Function”)

Event Attributes

Attribute Name	Opt./ Req.	<i>Notes</i>
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Event Attributes		
Name	R	Name
Description/Definition	R	Description
Processing Code	O	Type (Simple, Nesting, Concurrent Superstate if applicable)

Relationship Type Attributes

Attribute Name	Opt./ Req.	<i>Notes</i>
Relationship Type Attributes		
Name	R	Name
Description/Definition	R	Description
Processing Code	O	Type (Simple, Nesting, Concurrent Superstate if applicable)

Entity Type Attributes

Attribute Name	Opt./ Req.	<i>Notes</i>
Entity Type Attributes		
Name	R	Name

Description/Definition	R	Description
Identifier	O	Identifier
Processing Code	R	Transaction Type
Short Description	O	Format Type
Application	O	Media Type

Generalization Type Attributes

Attribute Name	Opt./ Req.	<i>Notes</i>
Generalization Type Attributes		
Name	R	Name
Description/Definition	R	Description

D Attribute Attributes

Attribute Name	Opt./ Req.	<i>Notes</i>
D Attribute Attributes		
Name	R	Name
Description/Definition	R	Description

Reference	O	Source
Type	O	Role of attribute, For example: Key, Foreign Key, Non-Key

Application System Attributes

Attribute Name	Opt./ Req.	<i>Notes</i>
Application System Attributes		
Name	R	Name
Description/Definition	R	Description
Short Description	O	Type
Source	O	Vendor/Source
Database Management System	O	Data Repository Name

Network Node Attributes

Attribute Name	Opt./ Req.	<i>Notes</i>
Network Node Attributes		

Name	R	Name
Description/Definition	R	Description
Short Description	O	Type
Source	O	Vendor/Source

Network Connection Attributes

Attribute Name	Opt./ Req.	Notes
Network Connection Attributes		
Name	R	Name
Description/Definition	R	Communication Standard, Protocols Supported
Transmission Speed	O	Capacity
Transmission Medium	O	Infrastructure Technology
Number of Channels	O	Number of communication Links

Network Attributes

Attribute Name	Opt./ Req.	Notes

Network Attributes		
Name	R	Name
Description/Definition	R	Communication Standard, Protocols Supported
Processing code	O	Security Classification

Key performance Indicator Attributes

Attribute Name	Opt./ Req.	<i>Notes</i>
Key Performance Indicator Attributes		
Name	R	Name
Description/Definition	R	Communication Standard, Protocols Supported
User attribute Point in time	O	Timestamp
Processing Code	O	Measurement Time ID
Balanced Scorecard		
Actual Value	O	Expected Value

Objective Attributes

Attribute Name	Opt./ Req.	<i>Notes</i>
Objective Attributes		
Name	R	Name
Description/Definition	R	Communication Standard, Protocols Supported
User attribute Point in time	O	Timestamp
Processing Code	O	Measurement Time ID
Balanced Scorecard		
Plan Value	O	Measurement Value

References

C4ISR Architecture Framework, Version 1.0, 7 June 1996, and Version 2.0, 18 December 1997

Scheer, A.-W. Architecture of Integrated Information Systems: Business Process Modeling. Berlin: Springer-Verlag, 1999a.

Scheer, A.-W. Architecture of Integrated Information Systems: Business Process Frameworks. Berlin: Springer-Verlag, 1999b.

Scheer, A.-W.; Abolhassan, Ferri; Jost, Wolfram; Kirchmer, Mathias; Business Process Change Management: Springer-Verlag 2003

Spewak, Steven H. Enterprise Architecture Planning. New York: John Wiley & Sons, 1992

UNDER SECRETARY OF DEFENSE; Renaming and Update of the Financial Management Modernization Program (FMMP) to the Business Management Modernization Program (BMMP), Washington, DC, 2003

U.S. Department of Defense, C4ISR Architecture Framework, Version 2.0. Washington, DC, 1997.

U.S. Department of Defense, DoD Architecture Framework, Volume I, Version 1.0. Washington, DC, 2003.

U.S. Department of Defense, DoD Architecture Framework, Volume II, Version 1.0. Washington, DC, 2003.